

MiMo: Unlocking the Reasoning Potential of Language Model – From Pretraining to Posttraining

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Abstract

We present MiMo-7B, a large language model born for reasoning tasks, with optimization across both pre-training and post-training stages. During pre-training, we enhance the data preprocessing pipeline and employ a three-stage data mixing strategy to strengthen the base model’s reasoning potential. MiMo-7B-Base is pre-trained on 25 trillion tokens, with additional Multi-Token Prediction objective for enhanced performance and accelerated inference speed. During post-training, we curate a dataset of 130K verifiable mathematics and programming problems for reinforcement learning, integrating a test-difficulty-driven code-reward scheme to alleviate sparse-reward issues and employing strategic data resampling to stabilize training. Extensive evaluations show that MiMo-7B-Base possesses exceptional reasoning potential, outperforming even much larger 32B models. The final RL-tuned model, MiMo-7B-RL, achieves superior performance on mathematics, code and general reasoning tasks, surpassing the performance of OpenAI o1-mini. The model checkpoints are available at <https://github.com/xiaomimimo/MiMo>.

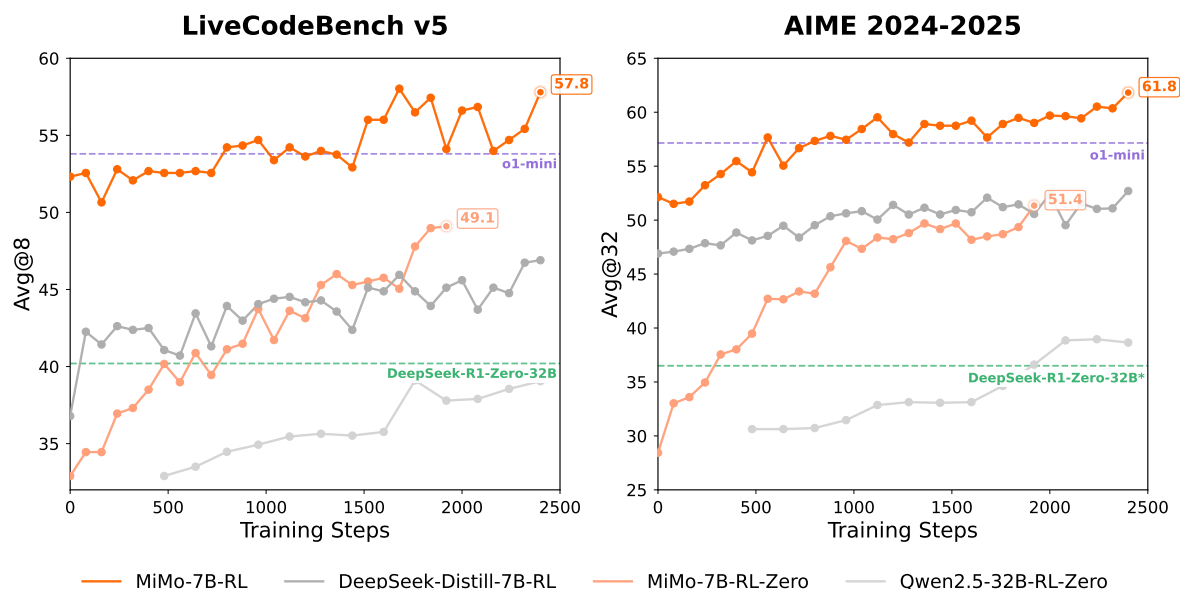


Figure 1 Performance of MiMo-7B in code and math reasoning benchmark.

MiMo：释放语言模型的推理潜力 – 从预训练到后训练

小米 LLM-Core 团队

抽象

我们介绍了 MiMo-7B，这是一种为推理任务而生的大型语言模型，在训练前和训练后阶段都进行了优化。在预训练期间，我们增强了数据预处理管道，并采用三阶段数据混合策略来增强基础模型的推理潜力。MiMo-7B-Base 在 25 万亿个代币上进行了预训练，并具有额外的 MultiToken 预测目标，以提高性能并加快推理速度。在训练后，我们策划了一个包含 130K 可验证数学和编程问题的数据集，用于强化学习，整合了测试难度驱动的代码奖励方案来缓解稀疏奖励问题，并采用战略数据重采样来稳定训练。广泛的评估表明，MiMo-7B-Base 具有非凡的推理潜力，性能甚至优于更大的 32B 模型。最终的 RL 调优模型 MiMo-7B-RL 在数学、代码和一般推理任务上实现了卓越的性能，超过了 OpenAI o1-mini 的性能。模型检查点可在 <https://github.com/xiaomimimo/MiMo> 获取。

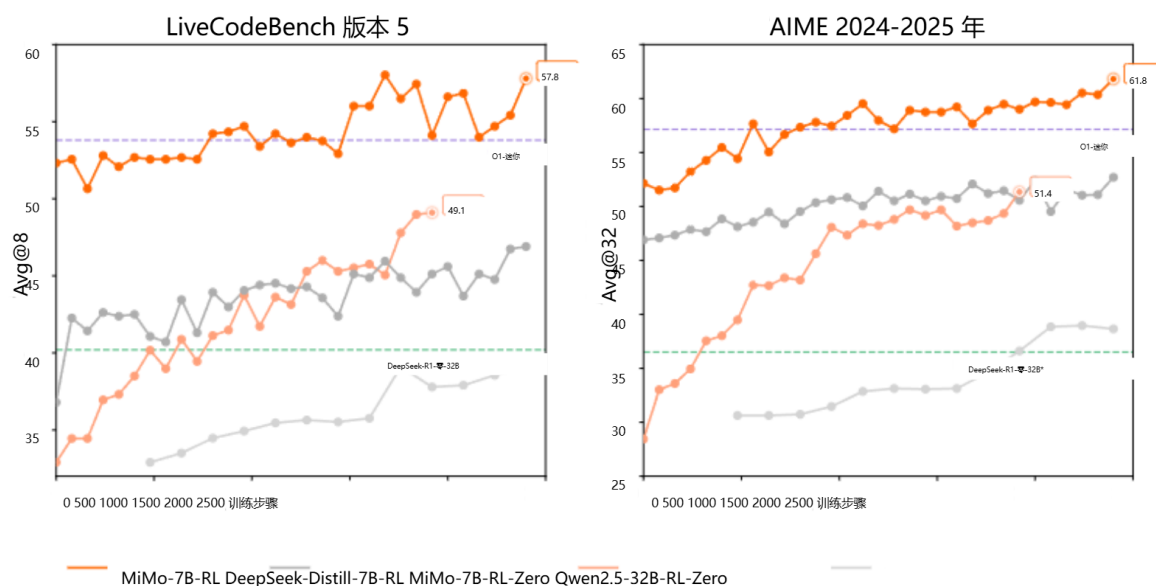


图 1 MiMo-7B 在代码和数学推理基准测试中的性能。

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1 Introduction

Large language models (LLMs) with advanced reasoning capabilities, such as OpenAI o-series (OpenAI, 2024), DeepSeek R1 (Guo et al., 2025), and Claude 3.7 (Anthropic, 2025), have achieved remarkable performance in complex tasks like mathematical reasoning and code generation. Through large-scale reinforcement learning (RL), these models develop sophisticated reasoning patterns, including step-by-step analysis, self-reflection and backtracking, enabling more robust and accurate problem solving capabilities across diverse domains. This emerging paradigm represents a significant advancement in artificial intelligence’s approach for tackling intricate challenges.

Currently, most successful RL works, including open-source research, rely on relatively large base models, e.g., 32B models, particularly for enhancing code reasoning capabilities. Moreover, it was widely considered that achieving uniform and simultaneous improvements in both mathematical and code capabilities within a small model is challenging. Nonetheless, we believe that the effectiveness of the RL trained reasoning model relies on the inherent reasoning potential of the base model. To fully unlock the reasoning potential of language models, efforts must focus not only on post-training but also on pre-training strategies tailored to reasoning.

In this work, we present MiMo-7B, a series of models trained from scratch and born for reasoning tasks. Our RL experiments from MiMo-7B-Base show that our model possesses extraordinary reasoning potential, even outperforming much larger 32B models. Additionally, we perform RL training on a cold-started SFT model, resulting in MiMo-7B-RL, which demonstrates superior performance on both mathematics and code reasoning tasks, surpassing the performance of OpenAI o1-mini. Here are our detailed contributions:

Pre-Training: Base Model Born for Reasoning

- We optimize data preprocessing pipeline, enhancing text extraction toolkits and applying multi-dimensional data filtering to increase reasoning pattern density in pre-training data. We also employ multiple strategies to generate massive diverse synthetic reasoning data.
- We adopt a three-stage data mixture strategy for pre-training. Overall, MiMo-7B-Base is pre-trained on approximately 25 trillion tokens.
- We incorporate Multiple-Token Prediction as an additional training objective, which enhances model performance and accelerates inference.

Post-Training Recipe: Pioneering Reasoning Model

- We curate 130K mathematics and code problems as RL training data, which can be verified by rule-based verifiers. Each problem undergoes careful cleaning and difficulty assessment to ensure quality. We employ only rule-based accuracy rewards to avoid potential reward hacking.
- To mitigate the sparse reward issue for challenging code problems, we introduce a test difficulty driven code reward. By assigning fine-grained scores for test cases with varying difficulty levels, the policy can be more effectively optimized via dense reward signal.
- We implement a data re-sampling strategy to enhance rollout sampling efficiency and stabilize policy updates, particularly in the later phases of RL training.

1 引言

具有高级推理功能的大型语言模型（LLM），例如 OpenAI o 系列（OpenAI, 2024 年）、DeepSeek R1（Guo 等人, 2025 年）和 Claude 3.7（Anthropic, 2025 年），在数学推理和代码生成等复杂任务中取得了显著的性能。通过大规模强化学习（RL），这些模型开发了复杂的推理模式，包括逐步分析、自我反思和回溯，从而在不同领域实现更强大、更准确的问题解决能力。这种新兴的范式代表了人工智能应对复杂挑战的方法的重大进步。

目前，大多数成功的 RL 工作，包括开源研究，都依赖于相对较大的基础模型，例如 32B 模型，特别是用于增强代码推理能力。此外，人们普遍认为，在小型模型中实现数学和代码功能的统一和同步改进是具有挑战性的。尽管如此，我们认为 RL 训练推理模型的有效性取决于基础模型固有的推理潜力。为了充分释放语言模型的推理潜力，工作不仅必须集中在训练后，还必须关注为推理量身定制的训练前策略。

在这项工作中，我们介绍了 MiMo-7B，这是一系列从头开始训练并为推理任务而生的模型。我们在 MiMo-7B-Base 的 RL 实验表明，我们的模型具有非凡的推理潜力，甚至超过了更大的 32B 模型。此外，我们在冷启动的 SFT 模型上执行 RL 训练，产生了 MiMo-7B-RL，它在数学和代码推理任务上都表现出卓越的性能，超过了 OpenAI o1-mini 的性能。以下是我们的详细贡献：

预训练：为推理而生的基础模型

- 我们优化了数据预处理管道，增强了文本提取工具包，并应用了多维数据过滤来提高预训练数据中的推理模式密度。
我们还采用多种策略来生成大量不同的综合推理数据。
- 我们采用三阶段数据混合策略进行预训练。总体而言，MiMo-7B-Base 在大约 25 万亿个代币上进行了预训练。
- 我们将多标记预测作为额外的训练目标，从而提高模型性能并加速推理。

训练后秘诀：开创性推理模型

- 我们将 130K 数学和代码问题作为 RL 训练数据进行管理，这些数据可以由基于规则的验证器进行验证。每个问题都经过仔细的清洁和难度评估，以确保质量。我们只使用基于规则的准确性奖励，以避免潜在的奖励黑客攻击。
- 为了缓解具有挑战性的代码问题的稀疏奖励问题，我们引入了测试难度驱动的代码奖励。通过为不同难度级别的测试用例分配细粒度的分数，可以通过密集的信号更有效地优化策略。
- 我们实施了数据重采样策略，以提高推出采样效率并稳定策略更新，尤其是在 RL 训练的后期阶段。

RL Infrastructures

- We develop a Seamless Rollout Engine to accelerate RL training and validation. Our design integrates continuous rollout, asynchronous reward computation, and early termination to minimize GPU idle time, achieving $2.29\times$ faster training and $1.96\times$ faster validation.
- We support MTP in vLLM and enhance the robustness of the inference engine in RL system.

Summary of Evaluation Results

- **MiMo-7B-Base** outperforms SoTA open-source models of approximately 7B parameters, excelling in general knowledge and coding tasks. On BBH, it achieves a score of 75.2, showcasing superior reasoning capabilities. Its strong performance on SuperGPQA further highlights its ability to handle complex graduate-level questions.
- **MiMo-7B-RL-Zero** surpasses the RL training performance of the 32B base model on both mathematics and code tasks. This underscores its efficiency and potential in RL training, positioning MiMo-7B as a compelling candidate for future advancements in RL.
- **MiMo-7B-RL** achieves excellent reasoning performance. It scores 55.4 on AIME 2025, exceeding o1-mini by 4.7 points. In algorithm code generation tasks, MiMo-7B-RL demonstrates extremely impressive results, significantly outperforming OpenAI o1-mini on both LiveCodeBench v5 and the latest v6, demonstrating robust and stable capabilities. MiMo-7B-RL also maintains competitive general performance.

Open-Source We open-source MiMo-7B series, including checkpoints of the base model, SFT model, RL model trained from base model, and RL model trained from the SFT model. We believe this report along with the models will provide valuable insights to develop powerful reasoning LLM that benefit the larger community.

2 Pre-Training

In this section, we first detail our strategies to enhance reasoning capabilities during MiMo-7B pre-training process, encompassing pre-training data construction, model architecture design, and hyper-parameter settings. Then we demonstrate the reasoning potential of MiMo-7B-Base model.

2.1 Pre-Training Data

The pre-training corpus for MiMo-7B integrates diverse sources, including web pages, academic papers, books, programming code, and synthetic data. We believe that incorporating more data with high-quality reasoning patterns during pre-training stage can substantially enhance the reasoning potential of the resulting language model. To achieve this goal, we first optimize our natural text preprocessing pipeline to improve quality and most importantly, reasoning data density. Second, we leverage advanced reasoning models to generate extensive synthetic reasoning data. Finally, we implement a three-stage data mixture strategy to maximize our model’s reasoning potential across various tasks and domains.

Better Reasoning Data Extraction Web pages naturally contain content with high density reasoning patterns, such as coding tutorial and mathematics blogs. However, we discover that commonly used extractors (Barbaresi, 2021) often fail to preserve mathematics equations and

RL 基础设施

- 我们开发了一个无缝部署引擎来加速 RL 训练和验证。我们的设计集成了持续推出、异步奖励计算和提前终止，以最大限度地减少 GPU 空闲时间，从而将训练速度提高 $2.29\times$ 倍，验证速度提高 $1.96\times$ 。
- 我们在 vLLM 中支持 MTP，并增强了 RL 系统中推理引擎的鲁棒性。

评估结果摘要

- MiMo-7B-Base 的性能优于具有约 7B 参数的 SoTA 开源模型，在常识和编码任务方面表现出色。在 BBH 上，它取得了 75.2 分，展示了卓越的推理能力。它在 SuperGPQA 上的出色表现进一步凸显了其处理复杂研究生问题的能力。
- MiMo-7B-RL-Zero 在数学和代码任务上都超过了 32B 基础模型的 RL 训练性能。这强调了它在 RL 训练中的效率和潜力，使 MiMo-7B 成为 RL 未来发展的令人信服的候选者。
- MiMo-7B-RL 实现了出色的推理性能。它在 AIME 55.4 上得分为 2025，比 o1-mini 高出 4.7 分。在算法代码生成任务中，MiMo-7B-RL 展示了令人印象深刻的结果，在 LiveCodeBench v5 和最新的 v6 上都明显优于 OpenAI o1-mini，展示了稳健和稳定的能力。MiMo-7B-RL 还保持了具有竞争力的一般性能。

开源 我们开源了 MiMo-7B 系列，包括基础模型的检查点、SFT 模型、从基础模型训练的 RL 模型和从 SFT 模型训练的 RL 模型。我们相信本报告以及这些模型将为开发强有力的推理 LLM 提供有价值的见解，从而使更大的社区受益。

2 训练前

在本节中，我们首先详细介绍了在 MiMo-7B 预训练过程中增强推理能力的策略，包括预训练数据构建、模型架构设计和超参数设置。然后，我们展示了 MiMo-7B-Base 模型的推理潜力。

2.1 预训练数据

MiMo-7B 的预训练语料库集成了各种来源，包括网页、学术论文、书籍、编程代码和合成数据。我们相信，在预训练阶段将更多数据与高质量的推理模式相结合，可以大大提高所得语言模型的推理潜力。为了实现这一目标，我们首先优化了我们的自然文本预处理管道，以提高质量，最重要的是，推理数据密度。其次，我们利用高级推理模型来生成广泛的综合推理数据。最后，我们实施了三阶段数据混合策略，以最大限度地提高模型在各种任务和领域的推理潜力。

更好的推理数据提取 网页自然包含具有高密度推理模式的内容，例如编码教程和数学博客。然而，我们发现常用的提取器（Barbaresi, 2021）通常无法保留数学方程式和

code snippets embedded in the webpage. To address this limitation, we develop a novel HTML-extraction tool specially optimized for mathematics content (Liu et al., 2024b; Paster et al., 2024; Zhou et al., 2025), code blocks, and forum websites. For papers and books, we enhance PDF parsing toolkits to better handle STEM and code content. With these optimized extraction tools, we successfully preserved massive reasoning patterns for subsequent processing stages.

Fast Global Deduplication Data deduplication plays an important role in improving training efficiency and reducing overfitting. We adopt both URL deduplication and MinHash deduplication (Broder, 1997) across all webpage dumps. Through extreme engineering optimization, we can complete this global deduplication process within a single day. Since deduplication algorithms treat high-quality and low-quality text equally without content awareness, we subsequently adjust the final data distribution according to multi-dimension quality scores.

Multi-Dimensional Data Filtering High-quality pre-training data with rich reasoning patterns is crucial for developing models with strong reasoning capabilities. We find that commonly used heuristic rule-based filters (Penedo et al., 2023, 2024) incorrectly filter high-quality web pages containing substantial mathematical and code content. To address this limitation, we instead fine-tune small LLMs to serve as data quality taggers, performing domain classification and multi-dimensional quality assessment.

Synthetic Reasoning Data Another crucial source for reasoning patterns is synthetic data generated by advanced reasoning models. We employ multiple strategies to generate diverse synthetic reasoning responses. First, we select STEM content tagged with high reasoning depth and prompt models to develop insightful analyses and perform in-depth thinking based on the source materials. Second, we gather mathematics and code problems and prompt reasoning models to solve them. Additionally, we incorporate general domain queries, particularly creative writing tasks. Notably, our preliminary experiments reveal that, unlike non-reasoning data, synthetic reasoning data can be trained for extremely high number of epochs without overfitting risk.

Three-Stage Data Mixture To optimize the pre-training data distribution, we adopt a three-stage data strategy in the final model training:

- **Stage 1:** We incorporate all data sources except synthetic responses for reasoning task queries. We downsample overrepresented content, such as ads, news, job postings, and materials with insufficient knowledge density and reasoning depth. We also upsample high-value data from professional domains with superior quality.
- **Stage 2:** Building on the curated distribution in Stage 1, we significantly increase mathematics and code related data to ~70% of the mixture. This approach is expected to enhance specialized skills without compromising general language abilities (Zhu et al., 2024). The first two stages are trained with an 8,192-token context length.
- **Stage 3:** To boost the capabilities for solving complex tasks, we further incorporate ~10% synthetic responses for mathematics, code, and creative writing queries. Simultaneously, we extend the context length from 8,192 to 32,768 in the final stage.

Through this process, we build a large high-quality pre-training dataset comprising approximately **25 trillion** tokens.

2021 年) 通常无法保留网页中嵌入的数学方程式和代码片段。为了解决这一限制, 我们开发了一种专门针对数学内容优化的新型 HTML 提取工具 (Liu et al., 2024b;Paster 等人, 2024 年;周 et al., 2025)、代码块和论坛网站。对于论文和书籍, 我们增强了 PDF 解析工具包, 以更好地处理 STEM 和代码内容。借助这些优化的提取工具, 我们成功地为后续处理阶段保留了大量推理模式。

快速全局去重 数据去重在提高训练效率和减少过拟合方面起着重要作用。我们在所有网页转储中都采用了 URL 重复数据删除和 MinHash 重复数据删除 (Broder, 1997)。通过极端的工程优化, 我们可以在一天内完成这个全局重复数据删除过程。由于去重算法在没有内容感知的情况下平等地对待高质量和低质量的文本, 因此我们随后根据多维度质量分数调整最终的数据分布。

多维度数据过滤 具有丰富推理模式的高质量预训练数据对于开发具有强大推理能力的模型至关重要。我们发现, 常用的基于规则的启发式过滤器 (Penedo et al., 2023, 2024) 会错误地过滤包含大量数学和代码内容的高质量网页。为了解决这一限制, 我们改为微调小型 LLM 以用作数据质量标记器, 执行域分类和多维质量评估。

合成推理数据 推理模式的另一个重要来源是由高级推理模型生成的合成数据。我们采用多种策略来生成不同的综合推理反应。首先, 我们选择标记有高推理深度的 STEM 内容, 并提示模型开发有洞察力的分析, 并根据源材料进行深入思考。其次, 我们收集数学和代码问题, 并提示推理模型来解决它们。此外, 我们还整合了一般的领域查询, 特别是创意写作任务。值得注意的是, 我们的初步实验表明, 与非推理数据不同, 合成推理数据可以训练极高数量的 epoch, 而不会有过拟合风险。

三阶段数据混合 为了优化预训练数据分布, 我们在最终的模型训练中采用了三阶段数据策略:

- 第 1 阶段: 我们整合了除用于推理任务查询的综合响应之外的所有数据源。我们对代表性过高的内容进行下采样, 例如广告、新闻、招聘启事以及知识密度和推理深度不足的材料。我们还对来自专业领域的高价值数据进行上采样, 质量上乘。
- 第 2 阶段: 在第 1 阶段的精选分发的基础上, 我们将数学和代码相关数据显著增加到混合物的 70% 左右。这种方法有望在不影响一般语言能力的情况下提高专业技能 (Zhu et al., 2024)。前两个阶段使用 8,192 个令牌的上下文长度进行训练。
- 第 3 阶段: 为了提高解决复杂任务的能力, 我们进一步为数学、代码和创意写作查询整合了 ~10% 的合成响应。同时, 我们在最后阶段将上下文长度从 8,192 扩展到 32,768。

通过这个过程, 我们构建了一个包含大约 25 万亿个代币的大型高质量预训练数据集。

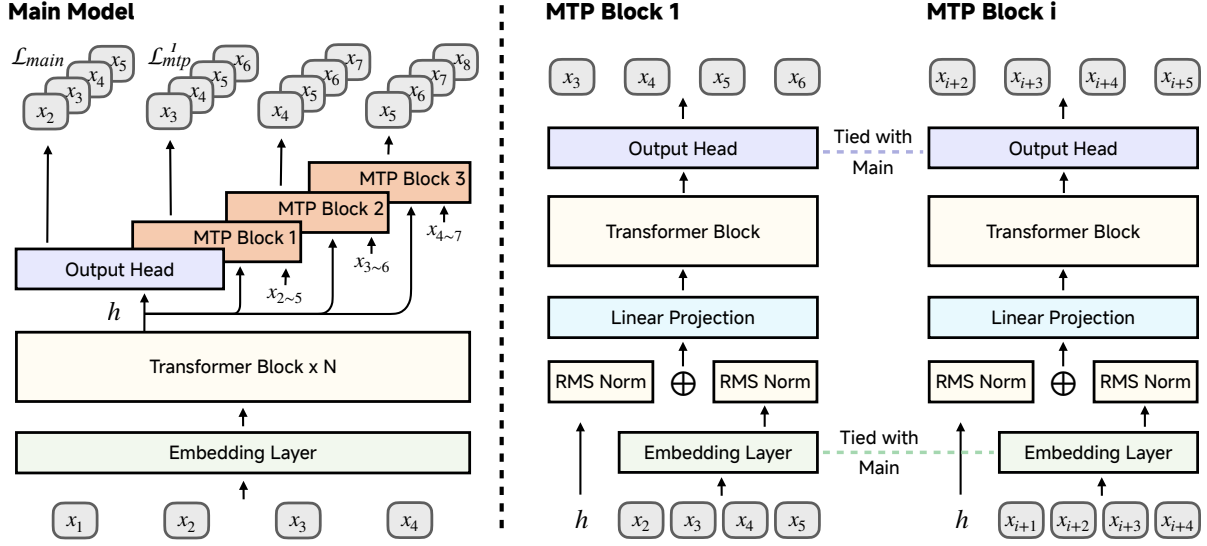


Figure 2 Implementation of Multi-Token Prediction with MiMo-7B. During pre-training we use a single MTP layer, while the inference stage can use multiple MTP layers for additional speedup.

2.2 Model Architecture

MiMo-7B follows the general decoder-only Transformer architecture (Radford et al., 2018; Vaswani et al., 2017), and consists of Grouped-Query Attention (GQA, Ainslie et al. 2023), pre-RMSNorm (Zhang and Sennrich, 2019), SwiGLU activation (Dauphin et al., 2017) and Rotary Positional Embedding (RoPE, Su et al. 2024), similar to Llama (Grattafiori et al., 2024; Touvron et al., 2023) and Qwen (Yang et al., 2024).

Reasoning models often face an inference speed bottleneck due to their lengthy auto-regressive generation process, despite the high correlation and predictability observed among consecutive tokens in their reasoning paths.

MTP Modules Inspired by DeepSeek-V3 (Liu et al., 2024a), we incorporate Multi-Token Prediction (MTP) (Gloeckle et al., 2024) as an additional training objective. This approach enables the model to strategically pre-plan and generate representations that facilitate more accurate and potentially faster prediction of future tokens. As shown in Figure 2, we implement distinct MTP setups for pre-training and inference. During pre-training, we utilize only a single MTP layer, as our preliminary studies show that multiple MTP layers yield no further improvement. In contrast, we find that multiple parallel MTP layers significantly accelerate inference through speculative decoding. To implement this, after pre-training, we replicate the pre-trained single MTP layer into two identical copies. Then, with the main model and first MTP layer frozen, we fine-tune two new MTP layers for inference speedup.

MTP Inference Speedup During inference, these MTP layers can be utilized for speculative decoding (Leviathan et al., 2023; Xia et al., 2023) to reduce generation latency. We evaluated the performance of the MTP layers on the AIME24 benchmark. The first MTP layer achieves a remarkably high acceptance rate about 90%, while even the third MTP layer maintains an acceptance rate above 75%. This high acceptance rate enables MiMo-7B to deliver enhanced decoding speed, particularly in reasoning scenarios requiring extremely long outputs.

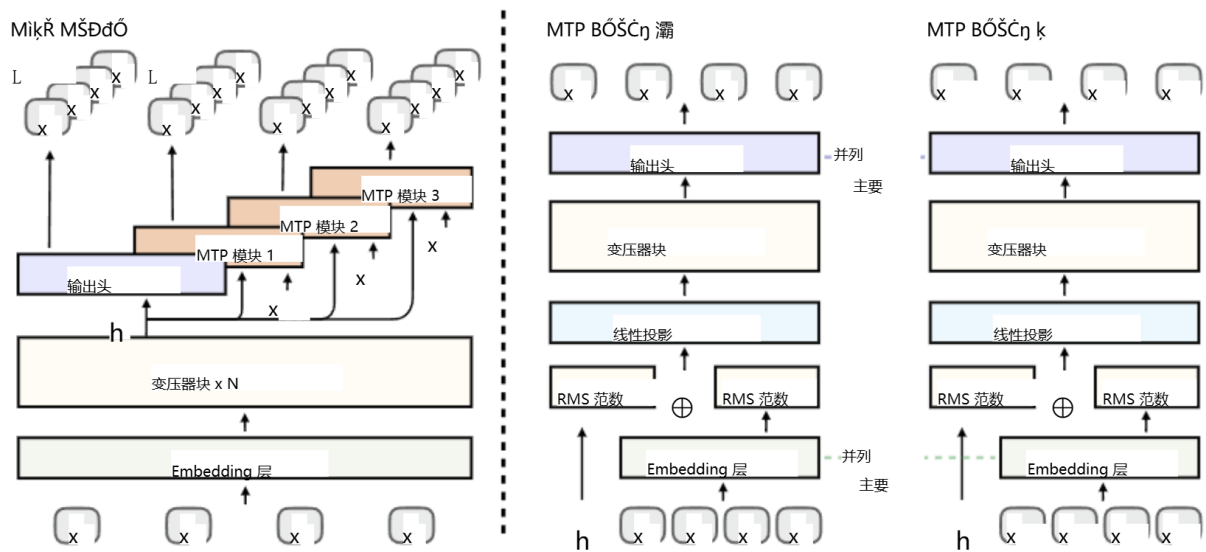


图 2：使用 MiMo-7B 实现多令牌预测。在预训练期间，我们使用单个 MTP 层，而推理阶段可以使用多个 MTP 层来进一步加速。

2.2 模型架构

MiMo-7B 遵循一般的仅解码器 Transformer 架构 (Radford 等人, 2018 年; Vaswani 等人, 2017 年), 由分组查询注意力 (GQA, Ainslie 等人, 2023 年)、前 RMSNorm (Zhang 和 Sennrich, 2019 年)、SwiGLU 激活 (Dauphin 等人, 2017 年) 和旋转位置嵌入 (RoPE, Su 等人, 2024 年) 组成, 类似于美洲驼 (Grattafiori 等人, 2024 年; Touvron 等人, 2023 年) 和 Qwen (Yang 等人, 2024 年)。

推理模型由于其漫长的自回归生成过程而经常面临推理速度瓶颈, 尽管在其推理路径中的连续标记之间观察到高度的相关性和可预测性。

MTP 模块受 DeepSeek-V3 (Liu et al., 2024a) 的启发, 我们将多标记预测 (MTP) (Gloeckle et al., 2024) 作为额外的训练目标。这种方法使模型能够战略性地预先规划和生成表示, 从而有助于更准确、更快速地预测未来代币。如图 2 所示, 我们实现了不同的 MTP 设置, 用于预训练和推理。在预训练期间, 我们只使用单个 MTP 层, 因为我们的初步研究表明, 多个 MTP 层不会产生进一步的改进。相比之下, 我们发现多个并行 MTP 层通过推测解码显著加速了推理。为了实现这一点, 在预训练之后, 我们将预训练的单个 MTP 层复制到两个相同的副本中。然后, 在冻结主模型和第一个 MTP 层的情况下, 我们微调两个新的 MTP 层以提高推理速度。

MTP 推理加速 在推理过程中, 这些 MTP 层可用于推测解码 (Leviathan 等人, 2023 年; Xia et al., 2023) 来减少生成延迟。我们在 AIME24 基准测试中评估了 MTP 层的性能。第一个 MTP 层实现了大约 90% 的非常高的接受率, 而即使是第三个 MTP 层也保持了 75% 以上的接受率。这种高接受率使 MiMo-7B 能够提供更高的解码速度, 尤其是在需要极长输出的推理场景中。

2.3 Hyper-Parameters

Model Hyper-Parameters We set the number of Transformer layers to 36 and the hidden dimension to 4,096. The intermediate hidden dimension of FFN is set to 11,008. The number of attention heads is 32 and there are 8 key-value groups.

Training Hyper-Parameters For optimization, we use AdamW (Loshchilov and Hutter, 2019) with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and weight decay of 0.1. We apply gradient clipping with a maximum norm of 1.0.

During the first two pre-training stages, the maximum sequence length is 8,192 tokens with the RoPE base of 10,000. Stage 3 expands these parameters to 32,768 tokens and 640,000, respectively.

Our learning rate schedule begins in Stage 1 with a linear warmup from 0 to 1.07×10^{-4} over the first 84B tokens, followed by a constant phase at 1.07×10^{-4} for 10.2T tokens, and concludes with a cosine decay to 3×10^{-5} over 7.5T tokens. This rate of 3×10^{-5} is maintained throughout Stage 2 (4T tokens) and for the first 1.5T tokens of Stage 3. Subsequently, the learning rate decays via a cosine schedule to 1×10^{-5} over the final 500B tokens.

We implement a linear batch size warmup to 2,560 over the first 168B tokens and maintain this value throughout the remainder of Stage 1 and Stage 2. In Stage 3, the batch size is fixed at 640.

The MTP loss weight is set to 0.3 for the first 10.3T tokens, then reduced to 0.1 for the remainder of pre-training.

2.4 Pre-Training Evaluation

2.4.1 Evaluation Setup

We evaluate MiMo-7B-Base on a series of benchmarks, encompassing natural language understanding and reasoning, scientific question answering, reading comprehension, mathematics reasoning, coding, Chinese understanding, and long-context comprehension capabilities:

Language understanding and reasoning: BBH (Suzgun et al., 2023), MMLU Hendrycks et al. (2021a), MMLU-Redux (Gema et al., 2024), MMLU-Pro (Wang et al., 2024), ARC (Clark et al., 2018), HellaSwag (Zellers et al., 2019), PIQA (Bisk et al., 2020).

Closed-book question answering: TriviaQA (Joshi et al., 2017), NaturalQuestions (Kwiatkowski et al., 2019).

Scientific question answering: GPQA (Rein et al., 2024), SuperGPQA (Du et al., 2025).

Reading comprehension: DROP (Dua et al., 2019), RACE (Lai et al., 2017).

Mathematics reasoning: AIME (MAA, 2024), GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021b).

Coding: LiveCodeBench (Jain et al., 2024), HumanEval (Chen et al., 2021), HumanEval+ (Liu et al., 2023), MBPP (Austin et al., 2021), MBPP+ (Liu et al., 2023), CRUXEval (Gu et al., 2024).

Miscellaneous: WinoGrande (Sakaguchi et al., 2020), AGIEval (Zhong et al., 2024a).

Chinese understanding: C-Eval (Huang et al., 2023), CMMLU (Li et al., 2023).

Long-Context Comprehension: RULER (Hsieh et al., 2024)

2.3 超参数

模型超参数我们将 Transformer 层数设置为 36，隐藏维度设置为 4,096。FFN 的中间隐藏维度设置为 11,008。注意力头的数量为 32 个，有 8 个键值组。

训练超参数 为了优化，我们使用 AdamW (Loshchilov 和 Hutter, 2019)， $\beta = 0.9$ ， $\beta = 0.95$ ，权重衰减为 0.1。我们应用最大范数为 1.0 的渐变裁剪。

在前两个预训练阶段，最大序列长度为 8,192 个令牌，RoPE 碱基为 10,000。第 3 阶段将这些参数分别扩展到 32,768 个代币和 640,000 个代币。

我们的学习率时间表从第 1 阶段开始，在前 84B 代币上从 0 到 1.07×10^{-5} 进行线性预热，然后是 10.2T 代币在 1.07×10^{-5} 的恒定阶段，最后以余弦衰减到 3×10^{-6} 结束，超过 7.5T 代币。这个 3×10^{-6} 的比率在整个阶段 2 (4T 代币) 和阶段 3 的前 1.5T 代币中保持不变。随后，学习率通过余弦计划衰减到最后 500B 令牌的 1×10^{-6} 。

我们在前 168B 令牌上实施线性批量大小预热到 2,560，并在第 1 阶段和第 2 阶段的剩余时间内保持此值。在阶段 3 中，批量大小固定为 640。

前 10.3T 令牌的 MTP 损失权重设置为 0.3，然后在剩余的预训练中减少到 0.1。

2.4 训练前评估

2.4.1 评估设置

我们根据一系列基准评估 MiMo-7B-Base，包括自然语言理解和推理、科学问答、阅读理解、数学推理、编码、中文理解和长语境理解能力：

语言理解和推理：BBH (Suzgun 等人, 2023 年)、MMLU Hendrycks 等人 (2021a)、MMLU-Redux (Gema 等人, 2024 年)、MMLU-Pro (Wang 等人, 2024 年)、ARC (Clark 等人, 2018 年)、HellaSwag (Zellers 等人, 2019 年)、PIQA (Bisk 等人, 2020 年)。

闭卷问答：TriviaQA (Joshi 等人, 2017 年)、NaturalQuestions (Kwiatkowski 等人, 2019 年)。

科学问答：GPQA (Rein 等人, 2024 年)、SuperGPQA (Du 等人, 2025 年)。

阅读理解：DROP (Dua et al., 2019)、RACE (Lai et al., 2017)。

数学推理：AIME (MAA, 2024)、GSM8K (Cobbe et al., 2021)、MATH (Hendrycks et al., 2021b)。

编码：LiveCodeBench (Jain 等人, 2024 年)、HumanEval (Chen 等人, 2021 年)、HumanEval+ (Liu 等人, 2023 年)、MBPP (Austin 等人, 2021 年)、MBPP+ (Liu 等人, 2023 年)、CRUXEval (Gu 等人, 2024 年)。

其他：WinoGrande (Sakaguchi 等人, 2020 年)、AGIEval (Zhong 等人, 2024a)。

中国理解：C-Eval (Huang et al., 2023)、CMMLU (Li et al., 2023)。

长语境理解：RULER (Hsieh et al., 2024)

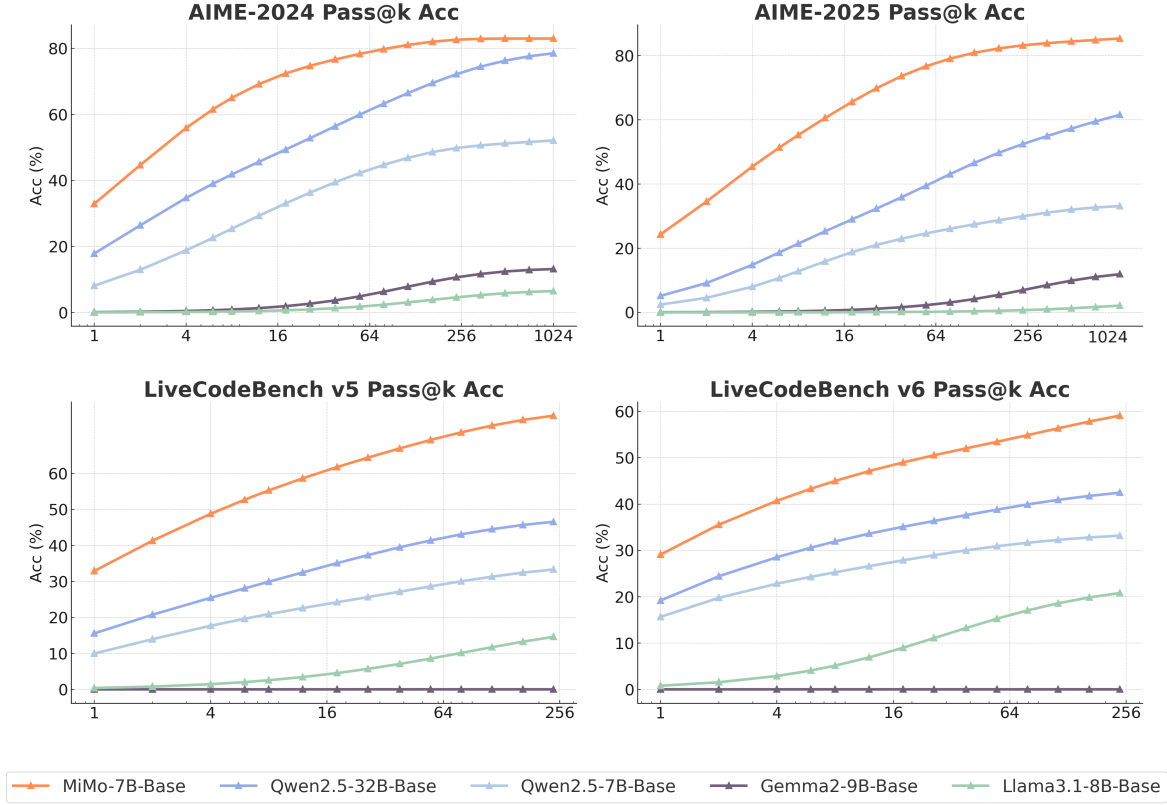


Figure 3 Pass@k curves of different base models across multiple reasoning benchmarks.

We compare MiMo-7B-Base with other open-source base models of comparable size, including Llama-3.1-8B (Grattafiori et al., 2024), Gemma-2-9B (Team, 2024), and Qwen2.5-7B (Yang et al., 2024). The evaluation of all models shares the same evaluation settings.

2.4.2 Upper Bounds of Reasoning Capability

Traditional evaluation methods often underestimate a model’s true reasoning potential by relying on single-pass success rates or average performance across multiple samplings. Following Yue et al. (2025), we adopt the pass@k metric, which considers a problem solved if any of k sampled solution is correct, to better assess the reasoning capacity boundary of different models.

As illustrated in Figure 3, MiMo-7B-Base achieves significantly higher pass@k scores than all compared models, including the 32B baseline, across all benchmarks and evaluated k values. Notably, the performance gap between MiMo-7B-Base and other baselines widens steadily as k increases, particularly on LiveCodeBench. These results demonstrates the superior reasoning potential of MiMo-7B-Base, which establishes a strong base policy for RL training.

2.4.3 Evaluation Results

General Reasoning MiMo-7B-Base achieves superior performance in general knowledge and reasoning, outperforming open-source models of comparable size. On BBH, a benchmark evaluating language reasoning abilities, MiMo-7B-Base scores 75.2, surpassing Qwen2.5-7B by about 5 points. Furthermore, SuperGPQA results show our model’s robust performance in solving graduate-level problems. On DROP, a reading comprehension benchmark, MiMo-7B-Base outperforms compared

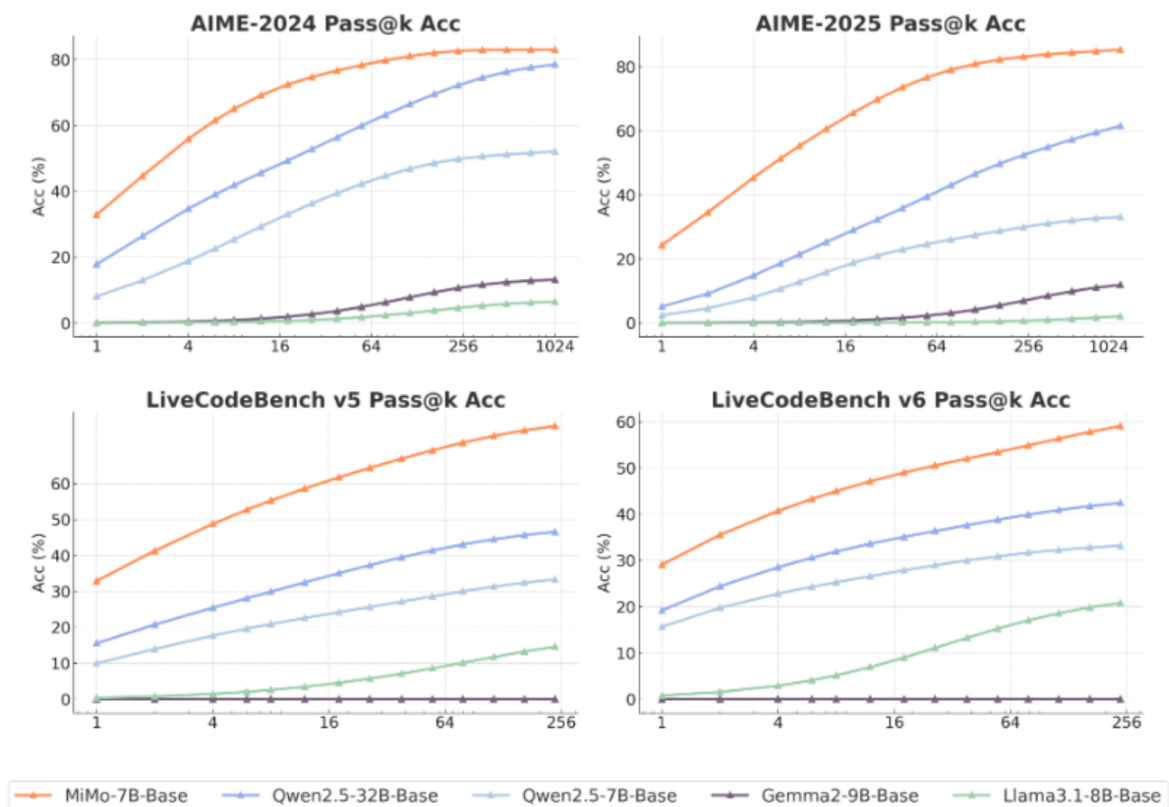


图 3 Pass@k 跨多个推理基准的不同基础模型的曲线。

2024 年) 我们将 MiMo-7B-Base 与其他类似大小的开源基础模型进行了比较, 包括 Llama-3.1-8B (Grattafiori 等人, 2024 年)、Gemma-2-9B (团队, 2024 年) 和 Qwen2.5-7B (Yang 等人, 2024 年)。所有模型的评估共享相同的评估设置

2.4.2 推理能力上限

传统的评估方法通常依赖于单次通过的成功率或多次采样的平均性能, 从而低估了模型的真实推理潜力。遵循 Yue et al. (2025), 我们采用 pass@k 度量, 如果 k 个采样解决方案中的任何一个是正确的, 则认为问题已解决, 以更好地评估不同模型的推理能力边界。

如图 3 所示, MiMo-7B-Base 在所有基准和评估的 k 值中都获得了明显高于所有比较模型 (包括 32B 基线) 的 pass@k 分数。值得注意的是, 随着 k 的增加, MiMo-7B-Base 和其他基线之间的性能差距稳步扩大, 尤其是在 LiveCodeBench 上。这些结果证明了 MiMo-7B-Base 的卓越推理潜力, 它为 RL 训练建立了强大的基础策略。

2.4.3 评估结果

一般推理 MiMo-7B-Base 在一般知识和推理方面取得了卓越的性能, 优于同等规模的开源模型。在评估语言推理能力的基准测试 BBH 上, MiMo-7B-Base 得分为 75.2, 比 Qwen2.5-7B 高出约 5 分。此外, SuperGPQA 结果表明, 我们的模型在解决研究生水平的问题方面具有稳健的性能。在阅读理解基准 DROP 上, MiMo-7B-Base 的表现优于

Benchmark	# Shots	Llama-3.1 8B Base	Gemma-2 9B Base	Qwen2.5 7B Base	MiMo- 7B Base
General					
BBH _(EM)	3-shot	64.2	69.4	70.4	75.2
GPQA-Diamond _(EM)	5-shot	33.3	24.2	35.4	25.8
SuperGPQA _(EM)	5-shot	19.9*	22.6*	24.6*	25.1
DROP _(F1)	3-shot	59.5	67.9*	61.5*	69.2
MMLU _(EM)	5-shot	65.3	71.2	74.2	71.2
MMLU-Redux _(EM)	5-shot	58.4*	67.9	71.1	65.3
MMLU-Pro _(EM)	5-shot	37.1	44.7	45.0	41.9
ARC-Easy _(EM)	25-shot	84.3	88.3	86.4	85.2
ARC-Challenge _(EM)	25-shot	57.7	68.2	63.8	62.3
HellaSwag _(EM)	10-shot	82.0	81.9	80.4	80.0
PIQA _(EM)	0-shot	80.3	81.9	78.5	79.4
WinoGrande _(EM)	5-shot	60.5	73.9*	75.9	78.0
RACE-High _(EM)	5-shot	44.3	48.3	46.8	44.1
TriviaQA _(EM)	5-shot	70.6	76.5	60.0	60.8
NaturalQuestions _(EM)	5-shot	27.7	29.2	24.1	24.5
AGIEval _(EM)	0-shot	38.2*	21.6*	44.4	48.3
Mathematics					
AIME 2024 _(Pass@1)	0-shot	0.3*	0.0*	10.1*	32.9
AIME 2025 _(Pass@1)	0-shot	0.0*	0.0*	4.3*	24.3
GSM8K _(EM)	8-shot	48.5*	70.2*	80.2*	75.2
MATH _(EM)	4-shot	16.9*	36.4*	44.3*	37.4
Code					
LiveCodeBench v5 _(Pass@1)	0-shot	0.4*	0.0*	5.0*	32.9
HumanEval _(Pass@1)	1-shot	37.8*	41.5*	56.7*	51.8
HumanEval+ _(Pass@1)	1-shot	31.7*	31.1*	50.0*	44.5
MBPP _(Pass@1)	3-shot	58.4	63.9	76.7	69.2
MBPP+ _(Pass@1)	3-shot	49.9	52.9	64.2	56.6
CRUXEval-I _(EM)	2-shot	41.5	49.8	52.4	47.6
CRUXEval-O _(EM)	2-shot	36.8	42.4	48.5	56.3
Chinese					
C-Eval _(EM)	5-shot	52.2	57.0	81.8	68.7
CMMLU _(EM)	5-shot	52.1	58.4	82.7	70.9

Table 1 Comparison among MiMo-7B-Base and other open-source base models of comparable size. Results marked with * are obtained using our internal evaluation framework.

基准	# 镜头	Llama-3.1 底座 7B	Gemma-2 底座 7B	Qwen2.5 底座	MiMo8B 底座 9B
常规					
BBH 餐厅	3 镜头	64.2	69.4	70.4	75.2
GPQA-钻石	5 镜头	33.3	24.2	35.4	25.8
超级 GPQA	5 镜头	19.9	22.6	24.6	25.1
落	3 镜头	59.5	67.9	61.5	69.2
MMLU	5 镜头	65.3	71.2	74.2	71.2
MMLU-Redux	5 镜头	58.4	67.9	71.1	65.3
MMLU-Pro 系列	5 镜头	37.1	44.7	45.0	41.9
ARC-易	25 发	84.3	88.3	86.4	85.2
ARC 挑战赛	25 发	57.7	68.2	63.8	62.3
HellaSwag 赃物	10 次拍摄	82.0	81.9	80.4	80.0
皮卡	0 次射击	80.3	81.9	78.5	79.4
维诺格兰德	5 镜头	60.5	73.9	75.9	78.0
RACE-高	5 镜头	44.3	48.3	46.8	44.1
花絮 QA	5 镜头	70.6	76.5	60.0	60.8
自然问题	5 镜头	27.7	29.2	24.1	24.5
AGIEval	0 次射击	38.2	21.6	44.4	48.3
数学					
爱 2024	0 次射击	0.3	0.0	10.1	32.9
爱 2025	0 次射击	0.0	0.0	4.3	24.3
GSM8K 型	8 镜头	48.5	70.2	80.2	75.2
数学	4 发	16.9	36.4	44.3	37.4
法典					
LiveCodeBench v5 0 镜头		0.4	0.0	5.0	32.9
HumanEval	1 次	37.8	41.5	56.7	51.8
HumanEval+	1 次	31.7	31.1	50.0	44.5
MBPP	3 镜头	58.4	63.9	76.7	69.2
MBPP+	3 镜头	49.9	52.9	64.2	56.6
CRUXEval-I	2 次射击	41.5	49.8	52.4	47.6
CRUXEval-O	2 次射击	36.8	42.4	48.5	56.3
中文					
C-评估	5 镜头	52.2	57.0	81.8	68.7
CMMLU 公司	5 镜头	52.1	58.4	82.7	70.9

MiMo-7B-Base 的性能优于表 1 MiMo-7B-Base 与其他类似大小的开源基础模型之间的比较。标有 * 的结果使用我们的内部评估框架获得。

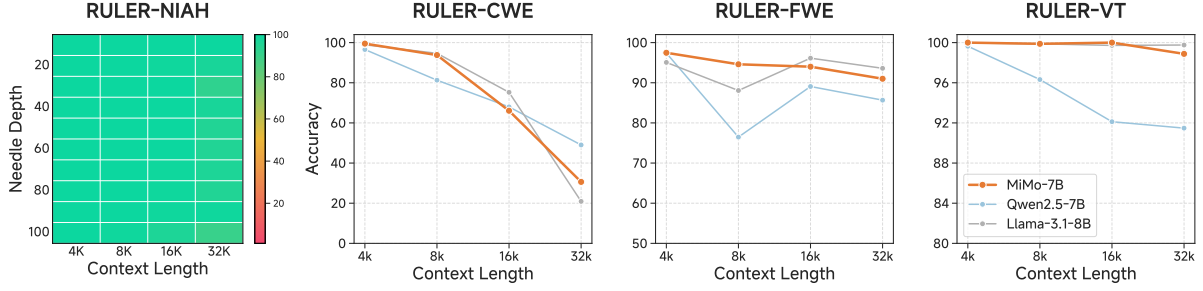


Figure 4 Results of long-context comprehension on RULER. Our MiMo-7B-Base achieves near-perfect NIAH retrieval performance within the supported 32K context length, and delivers remarkable performance on Common Words Extraction (CWE), Frequent Words Extraction (FWE), and Variable Tracking (VT) that emphasizes long-context reasoning beyond retrieval.

models, showing advanced language understanding capability.

Code and Mathematics Reasoning MiMo-7B-Base demonstrates strong proficiency in coding and mathematics tasks. On LiveCodeBench v5, it scores 32.9, far surpassing Llama-3.1-8B and Qwen-2.5-7B. Similarly, on AIME 2024, our model achieves 32.9, significantly outperforming other comparably sized base models. These results highlight MiMo-7B-Base’s extraordinary problem-solving abilities and its huge potential for complex reasoning tasks.

Long-Context Comprehension The ability to understand and reason over long contexts is essential for modern thinking models (Liu et al., 2025), as it enables them to produce long and complex reasoning chains.

For the needle-in-a-haystack (NIAH) tasks (Single, Multi-keys, Multi-values, and Multi-queries NIAH) that focus on long-context retrieval, we aggregate their accuracy across varying depths and context lengths, as depicted in the leftmost panel of Figure 4. We observe that MiMo-7B achieves near-perfect retrieval performance across all positions within the 32K context window.

Beyond pure retrieval, MiMo-7B excels in tasks requiring long-context reasoning, including Common Words Extraction (CWE), Frequent Words Extraction (FWE), and Variable Tracking (VT). It delivers remarkable performance and surpasses Qwen2.5-7B in most scenarios. These results validate the efficacy of our strategy to incorporate diverse data with high-quality reasoning patterns during pre-training.

3 Post-Training

After the pre-training stage, post-training are implemented on MiMo-7B-Base. Specifically, we develop MiMo-7B-RL-Zero through direct RL from MiMo-7B-Base, and MiMo-7B-RL trained from an SFT version of MiMo-7B.

3.1 Supervised Fine-Tuning

SFT Data The SFT data consists of a combination of open-source and proprietary distilled data. To ensure optimal quality and diversity, we implement a three stage preprocessing pipeline. First, we eliminate all training queries that have 16-gram overlap with evaluation benchmarks to prevent data leakage. Then, we exclude samples with mixing language or incomplete response. Finally, we

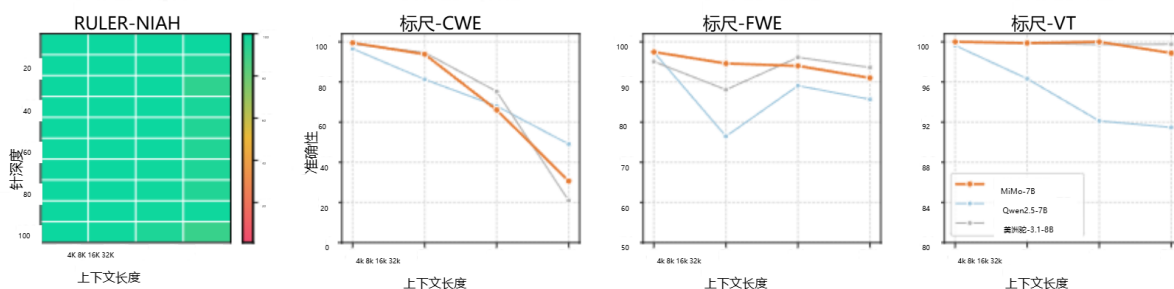


图 4 在 RULER 上进行长上下文理解的结果。我们的 MiMo-7B-Base 在支持的 32K 上下文长度内实现了近乎完美的 NIAH 检索性能，并在常用词提取（CWE）、高频词提取（FWE）和变量跟踪（VT）上表现出色，强调检索之外的长上下文推理。

模型，显示高级语言理解能力。

代码和数学推理 MiMo-7B-Base 在编码和数学任务方面表现出很强的熟练程度。在 LiveCodeBench v5 上，它的得分为 32.9，远高于 Llama-3.1-8B 和 Qwen-2.5-7B。同样，在 AIME 2024 上，我们的模型达到了 32.9，明显优于其他同等规模的基本模型。这些结果突出了 MiMo-7B-Base 非凡的问题解决能力及其在复杂推理任务中的巨大潜力。

长语境理解 在长语境中理解和推理的能力对于现代思维模型至关重要（Liu et al., 2025），因为它使它们能够产生长而复杂的推理链。

对于专注于长上下文检索的大海捞针（NIAH）任务（单、多键、多值和多查询 NIAH），我们汇总了它们在不同深度和上下文长度上的准确性，如图 4 最左侧的面板所示。我们观察到 MiMo-7B 在 32K 上下文窗口内的所有位置都实现了近乎完美的检索性能。

除了纯粹的检索之外，MiMo-7B 还擅长需要长上下文推理的任务，包括常用词提取（CWE）、高频词提取（FWE）和变量跟踪（VT）。它提供了卓越的性能，在大多数情况下都超过了 Qwen2.5-7B。这些结果验证了我们在预训练期间将不同数据与高质量推理模式相结合的策略的有效性。

3 培训后

在预训练阶段之后，在 MiMo-7B-Base 上实施后训练。具体来说，我们通过 MiMo-7B-Base 的直接 RL 开发 MiMo-7B-RL-Zero，并从 MiMo-7B 的 SFT 版本训练 MiMo-7B-RL。

3.1 监督微调

SFT 数据 SFT 数据由开源和专有提取数据的组合组成。为了确保最佳质量和多样性，我们实施了三个阶段预处理管道。首先，我们消除了所有与评估基准重叠 16 gram 的训练查询，以防止数据泄漏。然后，我们排除具有混合语言或不完全响应的样本。最后，我们

capped the number of responses per query at eight, striking a balance between preserving diversity and preventing redundancy. Following this preprocessing, our final SFT dataset comprises about 500K samples.

SFT Hyper-parameters We fine-tune the MiMo-7B-Base model with a constant learning rate of 3×10^{-5} and batch size of 128. Samples are packed to the maximum length of 32,768 tokens during training.

3.2 RL Data Curation

We utilize two categories of verifiable problems, mathematics and code, to formulate our RL training data. Our preliminary studies demonstrate that high-quality problem sets plays a critical role in stabilizing the RL training process and further enhancing the LLM’s reasoning capabilities.

Mathematical Data Our mathematical problem set is drawn from diverse sources, including open-source datasets and proprietary collected competition-level collections. To mitigate the risk of reward hacking, we employ an LLM to filter proof-based and multiple-choice problems. Unlike recent approaches that modify problems to ensure integer answers, we preserve original problems to minimize reward hacking. Additionally, we perform global n-gram deduplication and carefully decontaminate of our problem set with evaluation benchmarks.

Model-based difficulty assessment is used to further improve the quality of our dataset. Initially, we filter out problems that cannot be solved by advanced reasoning models, identifying those that are either too difficult or contain incorrect answers. For the remaining problems, we rollout an SFT version of MiMo-7B 16 times, eliminating problems with a passrate exceeding 90%. Notably, this process removes approximately 50% of easy problems from the original problem set. After data cleaning, we establish a mathematical training set comprising 100K problems.

Code Data For coding problems, we curate a high-quality training set comprising open-source datasets and our newly collected problem set. We remove problems without test cases. For problems with golden solutions, we exclude those where the golden solution failed to pass all test cases. For problems without golden solution, we discard problems where no test case can be solved in 16 rollouts of advanced reasoning models. Similar to math data, we utilize an SFT version of MiMo-7B to filter out easy problems that are perfectly solved in all 16 rollouts. This rigorous cleaning process yields 30K code problems.

During each RL iteration, we evaluate thousands of problems to compute the rewards, with each problem potentially containing hundreds of test cases. To improve reward computing efficiency and eliminate GPU idle time, we developed an online judge environment that enables parallel execution of extremely high-volume unit tests.

Reward Function We employ only rule-based accuracy rewards in our training process. For mathematics data, we use the rule-based Math-Verify library to evaluate response correctness. For code problems, we implement a test difficulty driven reward as detailed in Section 3.3.1. No additional rewards, such as format reward and length penalty reward, is incorporated.

3.3 RL Training Recipe

We employ a modified version of Group Relative Policy Optimization (GRPO) (Shao et al., 2024) with recently proposed improvement from the research community (Hu et al., 2025; Yu et al.,

我们将每个查询的响应数量限制为 8 个，在保持多样性和防止冗余之间取得平衡。经过此预处理，我们最终的 SFT 数据集包含大约 500K 样本。

SFT 超参数 我们以 3×10^{-5} 的恒定学习率和 128 的批量大小对 MiMo-7B-Base 模型进行微调。在训练期间，样本被打包到最大长度 32,768 个令牌。

3.2 RL 数据管理

我们利用两类可验证的问题，即数学和代码，来制定我们的 RL 训练数据。我们的初步研究表明，高质量的问题集在稳定 RL 训练过程和进一步增强 LLM 的推理能力方面起着关键作用。

数学数据 我们的数学问题集来自不同的来源，包括开源数据集和专有收集的比赛级集合。为了降低奖励黑客攻击的风险，我们采用 LLM 来过滤基于证明和多项选择的问题。与最近修改问题以确保整数答案的方法不同，我们保留了原始问题以最大限度地减少奖励黑客攻击。此外，我们执行全局 n -gram 重复数据删除，并使用评估基准仔细净化我们的问题集。

基于模型的难度评估用于进一步提高 out 数据集的质量。最初，我们过滤掉高级推理模型无法解决的问题，识别出那些太难或包含错误答案的问题。对于其余问题，我们推出了 MiMo-7B 的 SFT 版本 16 次，消除了通过率超过 90% 的问题。值得注意的是，此过程从原始问题集中删除了大约 50% 的简单问题。数据清洗后，我们建立了一个包含 100K 个问题的数学训练集。

代码数据 对于编码问题，我们策划了一个高质量的训练集，包括开源数据集和我们新收集的问题集。我们删除没有测试用例的问题。对于黄金解决方案的问题，我们排除了黄金解决方案未能通过所有测试用例的问题。对于没有黄金解决方案的问题，我们丢弃了在 16 次高级推理模型推出中无法解决的测试用例的问题。与数学数据类似，我们使用 MiMo-7B 的 SFT 版本来过滤掉在所有 16 次推出中完美解决的简单问题。这种严格的清理过程会产生 30K 代码问题。

在每次 RL 迭代期间，我们评估数千个问题以计算奖励，每个问题可能包含数百个测试用例。为了提高奖励计算效率并消除 GPU 空闲时间，我们开发了一个在线裁判环境，可以并行执行极大容量的单元测试。

奖励功能 我们在训练过程中只使用基于规则的准确性奖励。对于数学数据，我们使用基于规则的 Math-Verify 库来评估响应正确性。对于代码问题，我们实现了一个测试难度驱动奖励，如 Section 3.3.1 中所述。不包含其他奖励，例如格式奖励和长度惩罚奖励。

3.3 RL 训练配方

我们采用了组相对策略优化 (GRPO) 的修改版本 (Shao et al., 2024)，并结合了研究界最近提出的改进 (胡 et al., 2025; Yu et al.,

2025). For each problem q , the algorithm samples a group of responses $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{old}}$, and update the policy π_{θ} by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}_{q \sim D, \{o_i\}_{i=1}^G \sim \pi_{\theta}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{j=1}^{|o_i|} \min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_{i,j}, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \epsilon_{low}, 1 + \epsilon_{high} \right) A_{i,j} \right) \right] \quad (1)$$

where ϵ_{low} and ϵ_{high} are hyper-parameters. $A_{i,j}$ is the advantage, which is computed by the rewards $\{r_1, r_2, \dots, r_G\}$ of responses in the same group:

$$A_{i,j} = \frac{r_i - \text{mean}(\{r_i\}_{i=1}^G)}{\text{std}(\{r_i\}_{i=1}^G)} \quad (2)$$

Upon the original GRPO algorithm, we incorporate several enhancements from recent research:

- **Removal of KL Loss** (He et al., 2025; Hu et al., 2025): simply removing the KL loss effectively unleashes the full potential of the policy model without compromising training stability.
- **Dynamic Sampling** (Yu et al., 2025): in RL rollout phase, we over-sample and filter out prompts with passrate equal to 1 and 0, leaving all prompts in the batch with effective gradients while maintaining a consistent batch size. This strategy automatically calibrates problem difficulty throughout policy training.
- **Clip-Higher** (Yu et al., 2025): we increase the upper clip bounds ϵ_{high} in Eq. 1, with a fixed lower clip bounds ϵ_{low} . It can mitigate the entropy convergence problem and facilitate the policy to explore new solutions.

During training, we identify two key challenges affecting model performance: sparse rewards for code problems and diminishing sampling efficiency of dynamic sampling. Therefore, we propose **test complexity driven reward** function and **easy data re-sampling** approach, respectively.

3.3.1 Test Difficulty Driven Reward

Currently, for algorithm code generation tasks, existing RL works such as Deepseek-R1 Guo et al. (2025) adopt a rule-based reward strategy, where a solution is rewarded only if the generated code passes all the test cases for a given problem. However, for difficult algorithmic problems, the model might never receive any reward, preventing it from learning from these challenging cases and reducing training efficiency for dynamic sampling.

Various Test Difficulty in IOI Scoring Rules To address this limitation, we propose a new reward mechanism, test difficulty driven reward. The design is inspired by the scoring rule of the International Olympiad in Informatics (IOI, IOI 2024). In IOI contests, each complete problem is divided into multiple subtasks, and participants will obtain points for each subtask they successfully complete. Each subtask will have tests with different difficulty. Assigning different scores to subtasks better reflects how humans solve problems. For challenging problems, the model can still earn partial scores by solving some of the subtasks, which allows better utilization of these difficult examples during training.

Assigning Difficulty to Tests Based on Pass Rates We propose a technique for grouping test cases based on their difficulty. We utilize several models to perform multiple rollouts on each

2025 年)。对于每个问题 q ，算法从旧策略中抽取一组响应 $\{o_1, o_2, \dots, o_n\} \sim \pi_{\text{old}}$ ，并通过最大化以下目标来更新策略 π ：

$$J(\pi) = \mathbb{E} \left[\sum_{i=1}^n \frac{1}{|o_i|} \sum_{j=1}^{|o_i|} \frac{\pi(o_i|q) - \pi_{\text{old}}(o_i|q)}{\pi_{\text{old}}(o_i|q)} \frac{r_j - \text{avg}(\{r\})}{\text{std}(\{r\})} \right] \quad (1)$$

其中 ϵ 和 δ 是超参数。 A_i, j 是优势，它由同一组中响应的奖励 $\{r_1, r_2, \dots, r_n\}$ 计算得出：

$$A_i = \frac{r_i - \text{avg}(\{r\})}{\text{std}(\{r\})} \quad (2)$$

在原始 GRPO 算法的基础上，我们整合了最近研究的几项增强功能：

- 去除 KL 损失 (He et al., 2025; 胡 et al., 2025)：简单地消除 KL 损失可以有效地释放策略模型的全部潜力，而不会影响训练稳定性。
- 动态采样 (Yu et al., 2025)：在 RL 推出阶段，我们对通过率等于 1 和 0 的提示进行过度采样并过滤掉，使批次中的所有提示都具有有效的梯度，同时保持一致的批次大小。此策略在整个策略训练过程中自动校准问题难度。
- Clip-Higher (Yu et al., 2025)：我们在方程 1 中增加剪辑上限 ϵ ，并 δ 固定的剪辑下限。它可以缓解熵收敛问题，并促进政策探索新的解决方案。

在训练过程中，我们确定了影响模型性能的两个关键挑战：代码问题的稀疏奖励和动态采样采样效率的降低。因此，我们分别提出了测试复杂性驱动奖励函数和简单的数据重采样方法。

3.3.1 测试难度驱动奖励

目前，对于算法代码生成任务，现有的 RL 工作，如 Deepseek-R1 Guo et al. (2025) 采用基于规则的奖励策略，其中只有当生成的代码通过给定问题的所有测试用例时，解决方案才会获得奖励。然而，对于困难的算法问题，该模型可能永远不会获得任何奖励，从而阻止它从这些具有挑战性的案例中学习，并降低动态采样的训练效率。

IOI 评分规则中的各种测试难度 为了解决这一限制，我们提出了一种新的奖励机制，测试难度驱动的奖励。该设计的灵感来自国际信息学奥林匹克竞赛 (IOI, IOI 2024) 的评分规则。在 IOI 竞赛中，每个完成的问题被划分为多个子任务，参赛者成功完成每个子任务将获得积分。每个子任务都会有不同难度的测试。为子任务分配不同的分数可以更好地反映人类解决问题的方式。对于具有挑战性的问题，模型仍然可以通过解决一些子任务来获得部分分数，从而在训练过程中更好地利用这些困难的示例。

根据通过率为测试分配难度 我们提出了一种根据难度对测试用例进行分组的技术。我们利用多个模型在每个模型上执行多次部署

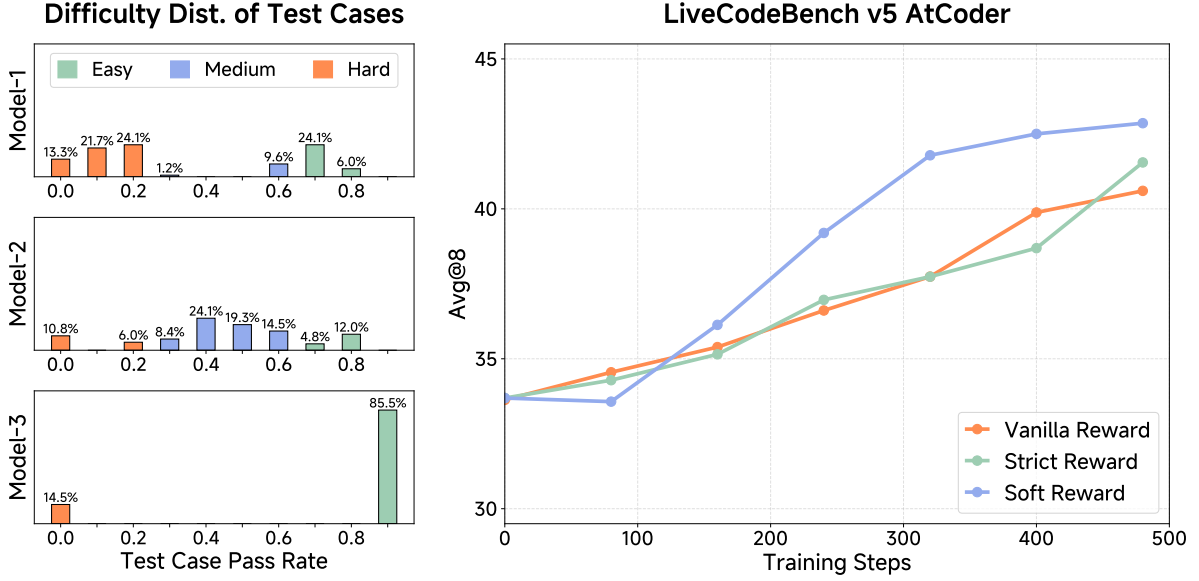


Figure 5 Experiments with test difficulty driven reward.

problem, and calculate the pass rate for each test case across all model-generated solutions. We then cluster the test cases into different difficulty levels according to their pass rates, with lower pass rates indicating higher difficulty. The left part of Figure 5 presents the pass rates and difficulty levels for each test case of certain problem. The results reveal a clear stratification of test difficulty, and demonstrate that more capable models achieve higher pass rates.

Reward Rules After categorizing the tests into different difficulty levels, we design two reward schemes based on these difficulty levels: a strict scheme and a soft scheme. (1) *Strict Reward*. Under the strict reward scheme, a solution receives the reward corresponding to a difficulty level only if it passes all tests in that group as well as in all lower-difficulty groups. (2) *Soft Reward*. In contrast, the soft reward scheme distributes the total score of each group equally among its tests. The final reward is the sum of the scores for all passed tests. The right part of Figure 5 compares the performance achieved by two reward schemes against the baseline without test difficulty driven reward.

3.3.2 Easy Data Filter and Re-Sampling

During RL training, as the policy improves, an increasing number of problems achieve a perfect pass rate of 1. Under dynamic sampling mechanism, these problems are subsequently filtered from the batch for policy update. This filtration leads to drastic sampling efficiency degradation, as more rollouts are required to construct a batch of fixed size. A straightforward approach to address this efficiency issue would be to entirely remove problems with perfect pass rates from the training data. However, our preliminary studies show that this method introduces significant instability in policy updates.

To improve sampling efficiency without risking policy collapse, we developed an easy data re-sampling strategy. During the training process, we maintain an easy data pool, where problems with perfect pass rates are stored. When performing rollouts, there is a probability α (10% in our experiments) to sample data from this easy data pool. This strategy effectively stabilizes the policy update while improving sampling efficiency, especially in the later phases of RL training.

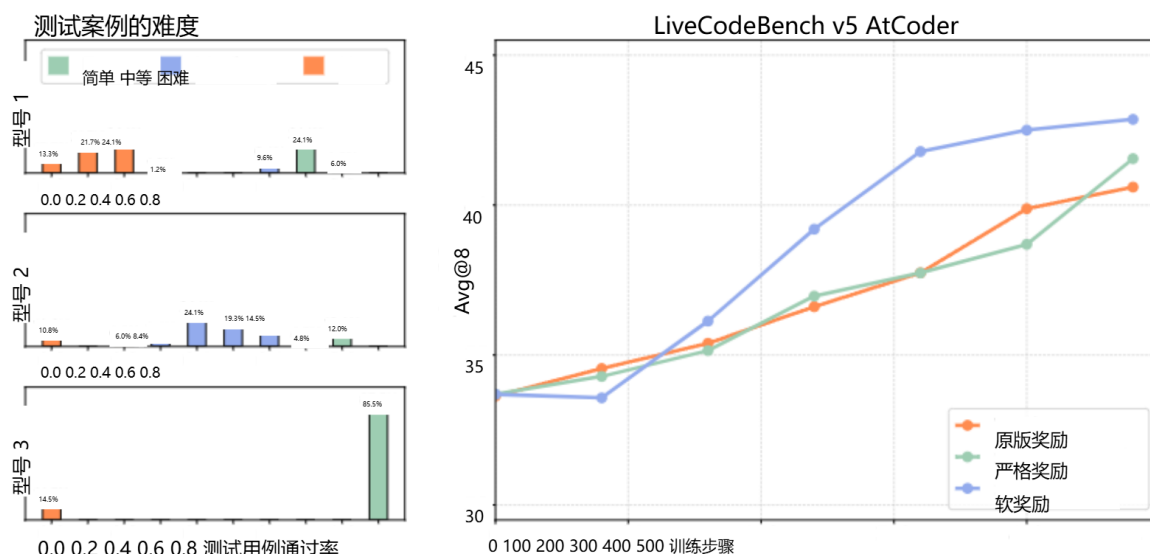


图 5 测试难度驱动奖励的实验。

我们利用多个模型对每个问题执行多次推出，并计算所有模型生成的解决方案中每个测试用例的通过率。然后，我们根据测试用例的通过率将测试用例分为不同的难度级别，通过率越低表示难度越高。图 5 的左侧部分显示了某个问题的每个测试用例的通过率和难度级别。结果揭示了测试难度的明确分层，并表明功能更强大的模型可实现更高的通过率。

奖励规则 在将测试分为不同的难度级别后，我们根据这些难度级别设计了两种奖励方案：严格方案和软方案。（1）严格奖励。在严格奖励方案下，只有当解决方案通过该组以及所有较低难度组的所有测试时，它才能获得与难度级别相对应的奖励。（2）软奖励。相比之下，软奖励方案在其测试中平均分配每组的总分。最终奖励是所有通过测试的分数总和。图 5 的右侧部分将两种奖励计划实现的性能与没有测试难度驱动奖励的基线进行了比较。

3.3.2 轻松的数据过滤和重新采样

在 RL 训练过程中，随着政策的完善，越来越多的问题达到了 1 的满分通过率。在动态采样机制下，这些问题随后会从批处理中筛选出来，以便进行策略更新。这种过滤会导致采样效率急剧下降，因为需要更多的推出来构建固定大小的批次。解决此效率问题的一种直接方法是从训练数据中完全消除完美通过率的问题。然而，我们的初步研究表明，这种方法在政策更新中引入了显著的不稳定性。

为了在不冒政策崩溃风险的情况下提高抽样效率，我们开发了一种简单的数据重采样策略。在培训过程中，我们维护一个简单的数据池，其中存储了完美通过率的问题。在执行部署时，有概率 α （在我们的实验中为 10%）从这个简单的数据池中采样数据。这种策略在提高采样效率的同时，有效地稳定了策略更新，尤其是在 RL 训练的后期。

3.3.3 Hyper-Parameters

In our experiment, we employed a training batch size of 512, with an actor mini-batch size of 32. We executed 16 gradient updates per training iteration at a learning rate of $1e-6$. The maximum sequence length was set to 32,768 tokens to facilitate complex reasoning tasks. During the training phase, both temperature and top-p parameters were configured at 1.0 to promote output diversity.

3.4 RL Infrastructures

We develop the Seamless Rollout Engine and enhance vLLM’s robustness to enable efficient dynamic-sampling-based RL training. We construct our RL system based on verl (Sheng et al., 2024), an open-source RL training library. The library uses Ray (Moritz et al., 2018) to manage computation and communication, implementing the rollout and training phases in Ray Actors and exchanging training data through Ray Objects. Although verl supports flexible implementations of various RL algorithms, it suffers from GPU idle time during both rollout and reward computation phases. Due to the skewness in response lengths, we observe that most GPUs remain idle while waiting for a few long-sequence rollout workers, resulting in wasted computational resources and a slow training process. Several prior works have identified this issue and proposed system-level solutions (Seed et al., 2025; Team et al., 2025; Zhong et al., 2024b). However, most of these solutions rely on asynchronous training, which modifies the underlying algorithm and introduces staleness in long-sequence responses. Rule-based reward computation is also time-consuming, particularly for code data, leading to idle periods for valuable GPU resources. Our use of dynamic sampling, while improving sample efficiency, exacerbates GPU idle time, and leads to wasted examples during multi-turn rollouts. To simultaneously optimize GPU utilization and reduce sample waste, we develop the Seamless Rollout Engine, opportunistically filling sample batches into rollout while performing asynchronous reward computation. Our system builds on the vLLM inference engine (Kwon et al., 2023), and we collaborate with the open-source community to enhance the robustness of vLLM’s “external launch” mode within the verl framework. Additionally, we implement MTP in vLLM to support both MiMo-7B and MiMo-7B-RL.

3.4.1 Seamless Rollout Engine

Seamless Rollout Engine optimizes GPU utilization in rollout workers through efficient task scheduling, minimizing idle time during continuous operation. The engine consists of the following components: (a) continuous rollout, (b) asynchronous reward computation, and (c) early termination. It achieves a $2.29\times$ speedup in training and a $1.96\times$ speedup in validation.

Continuous Rollout The core of Seamless Rollout Engine lies in proactively handling completed rollout tasks and initiating new rollouts. Unlike naive dynamic sampling implementations that delay reward computation until all rollout workers complete, Seamless Rollout Engine eliminates synchronization barriers between generation and reward phases. It actively monitors completed workers, immediately computes their rewards, and triggers new rollouts on demand. After computing rewards, we update the number of valid samples and the current step’s pass-rate statistics, then launch new rollout tasks if active tasks are insufficient to meet training demands based on these statistics. As illustrated in Figure 6, the Seamless Rollout Engine initiates a new task upon completing rollout tasks ③④①⑥ to meet demand, whereas after finishing tasks ②⑤⑦, it predicts that ongoing tasks are sufficient and thus schedules no additional ones.

3.3.3 超参数

在我们的实验中，我们采用了 512 的训练批次大小，其中 actor 小批量大小为 32。我们以 $1e-6$ 的学习率每次训练迭代执行 16 次梯度更新。最大序列长度设置为 32,768 个令牌，以方便执行复杂的推理任务。在训练阶段，temperature 和 top-p 参数都配置为 1.0 以促进输出多样性。

3.4 RL 基础设施

我们开发了 Seamless Rollout Engine 并增强了 vLLM 的稳健性，以实现基于动态采样的高效 RL 训练。我们基于开源 RL 训练库 verl (Sheng et al., 2024) 构建了我们的 RL 系统。该库使用 Ray (Moritz et al., 2018) 来管理计算和通信，在 Ray Actor 中实现推出和训练阶段，并通过 Ray 对象交换训练数据。尽管 verl 支持各种 RL 算法的灵活实现，但它在推出和奖励计算阶段都会受到 GPU 空闲时间的影响。由于响应长度的偏度，我们观察到大多数 GPU 在等待一些长序列推出工作程序时保持空闲状态，从而导致计算资源浪费和训练过程缓慢。之前的几项工作已经确定了这个问题并提出了系统级解决方案 (Seed 等人, 2025 年; Team et al., 2025; Zhong et al., 2024b)。但是，这些解决方案中的大多数都依赖于异步训练，这会修改底层算法并在长序列响应中引入过时性。基于规则的奖励计算也很耗时，尤其是对于代码数据，导致宝贵的 GPU 资源处于空闲期。我们使用动态采样，虽然提高了采样效率，但加剧了 GPU 空闲时间，并导致在多轮次部署期间浪费示例。为了同时优化 GPU 利用率并减少样本浪费，我们开发了无缝推出引擎，在执行异步奖励计算的同时，适时地将样本批次填充到推出中。我们的系统建立在 vLLM 推理引擎 (Kwon et al., 2023) 之上，并与开源社区合作，以增强 vLLM 的“外部启动”模式在 verl 框架内的稳健性。此外，我们在 vLLM 中实施了 MTP 以支持 MiMo-7B 和 MiMo-7B-RL。

3.4.1 无缝部署引擎

无缝部署引擎通过高效的调度优化部署工作程序中的 GPU 利用率，从而最大限度地减少连续运行期间的空闲时间。该引擎由以下组件组成：(a) 持续推出，(b) 异步奖励计算，以及 (c) 提前终止。它在训练中实现了 $2.29\times$ 的加速，在验证中实现了 $1.96\times$ 的加速。

持续推出无缝推出引擎的核心在于主动处理已完成的推出任务并启动新的推出。与将奖励计算延迟到所有推出工作程序完成的简单动态采样实现不同，无缝推出引擎消除了生成阶段和奖励阶段之间的同步障碍。它会主动监控已完成的工作程序，立即计算其奖励，并按需触发新的推出。计算奖励后，我们会更新有效样本的数量和当前步骤的通过率统计数据，然后根据这些统计数据启动新的推出任务（如果活动任务不足以满足训练需求）。如图 6 所示，无缝部署引擎在完成部署任务 (3) (4) (1) (6) 时启动新任务以满足需求，而在完成任务 (2) (5) (7) 后，它预测正在进行的任务就足够了，因此不会安排额外的任务。

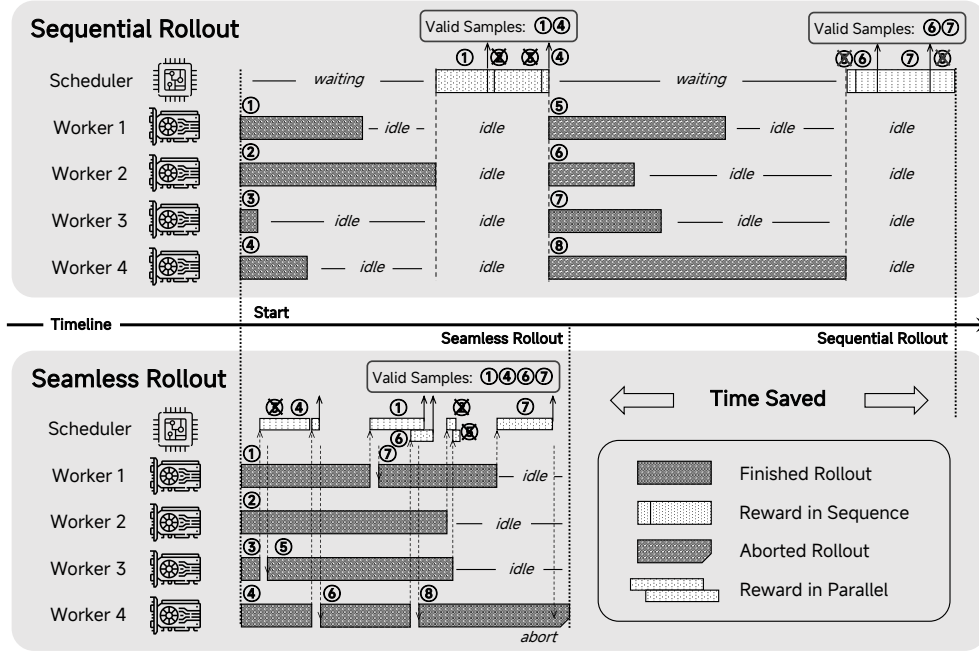


Figure 6 An overview of the Seamless Rollout Engine for MiMo-7B-RL.

Asynchronous Reward Computation While reward computation for math data is rapid, judging code-related data incurs significant overhead, leading to prolonged GPU idle time. Additionally, the sequential nature of naive reward computation fails to utilize the multiprocessing capabilities of modern processing units. To resolve these issues, we employ Ray to launch asynchronous reward computation, which facilitates concurrent management of rollout and reward tasks. Upon task completion, the system dynamically forwards rollout outputs for reward evaluation or aggregates results to update the sample state, as shown in Figure 6. Dedicated servers are allocated for code-specific reward computation to prevent bottlenecks in the rollout pipeline.

Early Termination When the number of valid samples exceeds the required training batch size, careful management of ongoing tasks becomes essential. Abrupt termination of ongoing tasks tends to suppress the generation of long-sequence responses, which could destabilize RL training dynamics. A straightforward solution involves waiting for all active tasks to complete before randomly sampling required batch from the outputs. However, this approach may extend waiting times if a long-sequence rollout initiates near the end of the dynamic sampling phase. To mitigate this delay while preserving data distribution integrity, we implement a first-in-first-out selection strategy. We terminate ongoing tasks only if the valid sample count meets the batch requirement and all tasks initiated prior to these selected samples have completed. In Figure 6, the last rollout is aborted since earlier samples already reach the required batch size.

Experimental Analysis We randomly choose a 5-step training trace to evaluate the performance of Seamless Rollout Engine. The experiment is conducted on 256 H20 GPUs, and the results are presented in Table 2. “Overall Speedup” measures end-to-end RL training efficiency; “Rollout Speedup” shows the acceleration of rollout and reward tasks; “Normalized GPU Idle Time” reflects the total idle GPU hours. The above metrics are normalized with respect to the naive dynamic sampling implementation. “GPU Idle Ratio” quantifies the average proportion of GPU inactivity during rollout and reward computation; “Sample Waste Ratio” represents the ratio of excess valid

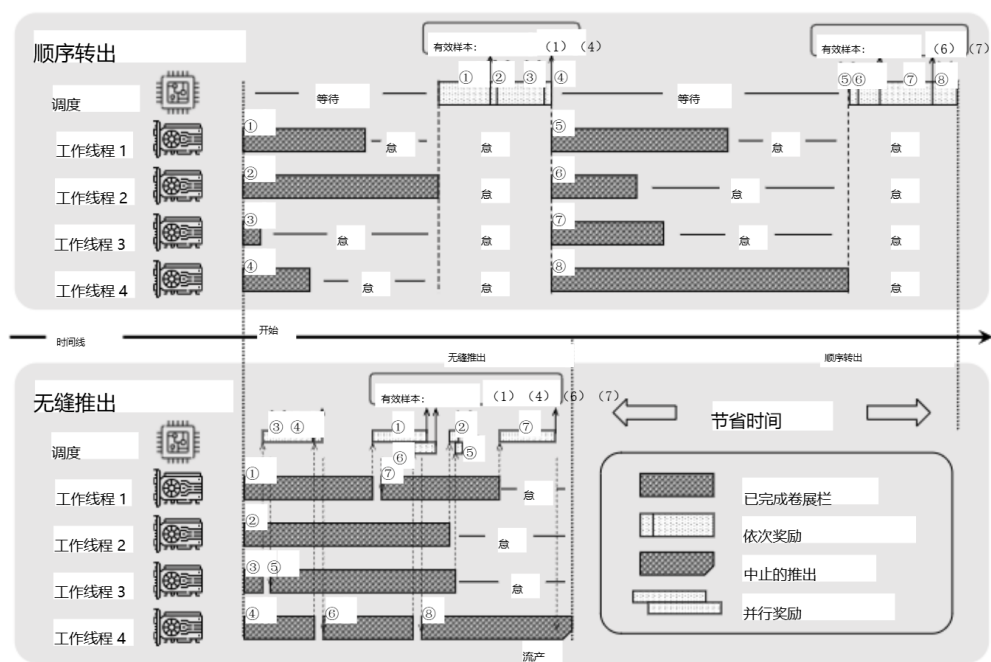


图 6 MiMo-7B-RL 的无缝部署引擎概述。

异步奖励计算虽然数学数据的奖励计算速度很快，但判断与代码相关的数据会产生大量开销，从而导致 GPU 空闲时间延长。此外，朴素奖励计算的顺序性无法利用现代处理单元的多处理功能。为了解决这些问题，我们使用 Ray 来启动异步奖励计算，这有助于并发管理上线和奖励任务。任务完成后，系统会动态转发推出输出以进行奖励评估，或聚合结果以更新示例状态，如图 6 所示。为特定于代码的奖励计算分配专用服务器，以防止推出管道中出现瓶颈。

提前终止 当有效样本的数量超过所需的训练批次大小时，仔细管理正在进行的任务变得至关重要。正在进行的任务的突然终止往往会抑制长序列响应的产生，这可能会破坏 RL 训练动力学的稳定性。一个简单的解决方案包括等待所有活动任务完成，然后再从输出中随机采样所需的批处理。但是，如果长序列转出在动态采样阶段接近结束时启动，则此方法可能会延长等待时间。为了在保持数据分发完整性的同时减少这种延迟，我们实施了先进先出的选择策略。只有当有效样品计数满足批次要求，并且在这些选定样品之前启动的所有任务都已完成时，我们才会终止正在进行的任务。在图 6 中，由于早期样本已经达到所需的批量大小，因此最后一次转出被中止。

实验分析：我们随机选择一个 5 步训练跟踪来评估 Seamless Rollout Engine 的性能。该实验在 256 个 H20 GPU 上进行，结果如表 2 所示。“整体加速比”衡量端到端的 RL 训练效率；“Rollout Speedup”展示 Rollout 和 Rewards 任务的加速；“标准化 GPU 空闲时间”反映的是 GPU 总空闲小时数。上述指标根据 naive dynamic sampling 实现进行了标准化。“GPU 空闲比率”量化了 GPU 在推出和奖励计算期间不活动的平均比例；“Sample Waste Ratio (样品废液比率)”表示有效超额比率

Method	Overall Speedup $\uparrow$	Rollout Speedup $\uparrow$	Normalized GPU Idle Time $\downarrow$	GPU Idle Ratio $\downarrow$	Sample Waste Ratio $\downarrow$
w/o Dynamic Sampling	2.45 $\times$	2.82 $\times$	0.36	70.8%	/
Naive Dynamic Sampling	1.00 $\times$	1.00 $\times$	1.00	69.3%	22.1%
+ Continuous Rollout	1.99 $\times$	2.20 $\times$	0.25	38.8%	13.9%
+ Async. Reward	2.09 $\times$	2.34 $\times$	0.21	34.0%	16.4%
+ Early Termination	2.29 $\times$	2.61 $\times$	0.15	27.7%	12.9%

Table 2 The experimental results of Seamless Rollout Engine compared with baseline methods.

samples generated relative to the required batch size. In Seamless Rollout Engine, aborted tasks are considered in GPU idle time.

All three components contribute to faster dynamic sampling and smaller GPU idle time. Though the experiment without dynamic sampling can achieve higher throughput, it incurs significant sample inefficiency due to numerous zero-gradient training samples. These zero-gradient samples not only diminishes the effective training batch size, but also risk destabilizing the training dynamics of the RL algorithm. Given an average sample pass rate of 41% within this 5-step experiment, static sampling achieves a sample efficiency similar to naive dynamic sampling; the latter does not train zero-gradient data but incurs wasted samples. Equipped with all three components, Seamless Rollout Engine achieves a comparable one-step training time compared to static sampling while demonstrating superior sample efficiency. The sample pass rate of 41% leads to a sample waste ratio of 22% in the naive implementation; in practice, this ratio can be larger in different situations. Through continuous rollout and dynamic launch scheduling, Seamless Rollout Engine reduces the sample waste ratio to around 15%.

Accelerated Validation During validation, we can directly stream the rollout and reward tasks using Seamless Rollout Engine. Similar to the naive implementation, currently we set the validation batch size equal to the dataset length and launch all rollout tasks simultaneously. Our implementation utilizes asynchronous reward computation, achieving a 1.96 $\times$ speedup while reducing idle GPU time to 25%, as demonstrated in Table 3. Notably, the experimental results demonstrate Seamless Rollout Engine’s potential for static sampling, which also has one-pass rollout and reward computation. If the validation dataset is sufficiently large, further acceleration can be achieved by optimizing the batch size for validation and employing continuous rollout.

Method	Speedup $\uparrow$	Normalized GPU Idle Time $\downarrow$	GPU Idle Ratio $\downarrow$
Naive Validation	1 $\times$	1	65.8%
Seamless Rollout Engine	1.96 $\times$	0.25	32.9%

Table 3 The validation speedup and GPU idle time of the naive implementation and the Seamless Rollout Engine. The experiment is conducted on 256 H20 GPUs using our full validation dataset.

3.4.2 vLLM-based Inference Engine

Our RL system employs vLLM (Kwon et al., 2023) as the inference engine. To accommodate our model’s new features, we have extended the framework with additional functionalities.

MTP Support As described in Section 2.2, our models integrate MTP modules to enhance performance. We have implemented and open-sourced MTP support for our models, enabling

方法	整体 加速 ↑	推出 加速 ↑	规范化 GPU 空闲时间 ↓	GPU 空闲 比率 ↓	样本 废弃率 ↓
无动态采样	2.45x	2.82x	0.36	70.8%	/
朴素动态采样 1.00x		1.00x	1.00	69.3%	22.1%
+ 持续推出	1.99x	2.20x	0.25	38.8%	13.9%
+ 异步。奖励	2.09x	2.34x	0.21	34.0%	16.4%
+ 提前终止	2.29x	2.61x	0.15	27.7%	12.9%

样本浪费率 “代表超额有效比率表 2 无缝推出引擎与基线方法的实验结果比较。

相对于所需批量大小生成的样本。在 Seamless Rollout Engine 中，中止的任务被视为 GPU 空闲时间。

这三个组件都有助于加快动态采样速度并缩短 GPU 空闲时间。虽然没有动态采样的实验可以实现更高的吞吐量，但由于大量的零梯度训练样本，它会导致严重的样本效率低下。这些零梯度样本不仅会减小有效的训练批次大小，而且还有可能破坏 RL 算法的训练动态。假设在这个 5 步实验中的平均样品通过率为 41%，静态采样可实现类似于朴素动态采样的样品效率；后者不训练零梯度数据，但会产生浪费的样本。Seamless Rollout Engine 配备了所有三个组件，与静态采样相比，可实现相当的一步训练时间，同时表现出卓越的采样效率。41% 的样本通过率导致 Naive 实施中样本浪费率为 22%；在实践中，这个比率在不同情况下可能会更大。通过持续推出和动态发布计划，无缝推出引擎将样品浪费率降低到 15% 左右。

加速验证 在验证过程中，我们可以使用 Seamless Rollout Engine 直接流式传输推出和奖励任务。与简单的实现类似，目前我们将验证 batch size 设置为等于数据集长度，并同时启动所有 rollout 任务。我们的实现利用异步奖励计算，实现了 1.96x 的加速，同时将空闲 GPU 时间减少到 25%，如表 3 所示。值得注意的是，实验结果表明 Seamless Rollout Engine 具有静态采样的潜力，该引擎还具有一次性推出和奖励计算功能。如果验证数据集足够大，则可以通过优化验证的批量大小并采用持续推出来实现进一步的加速。

方法	加速 ↑	标准化 GPU 空闲时间 ↓	GPU 空闲比率 ↓
朴素验证	1x	1	65.8%
无缝推出引擎 1.96x		0.25	32.9%

表 3 Naive 实现和 Seamless Rollout Engine 的验证加速和 GPU 空闲时间。该实验使用我们的完整验证数据集在 256 个 H20 GPU 上进行。

3.4.2 基于 vLLM 的推理引擎

我们的 RL 系统使用 vLLM (Kwon et al., 2023) 作为推理引擎。为了适应模型的新功能，我们扩展了框架，增加了一些功能。

MTP 支持如第 2.2 节所述，我们的模型集成了 MTP 模块以提高性能。我们已经为我们的模型实施并开源了 MTP 支持，从而支持

Benchmark	GPT-4o 0513	Claude-3.5- Sonnet-1022	OpenAI o1-mini	QwQ-32B Preview	R1-Distill- Qwen-14B	R1-Distill- Qwen-7B	MiMo- 7B-RL
General							
GPQA Diamond (Pass@1)	49.9	65.0	60.0	54.5	59.1	49.1	54.4
SuperGPQA (Pass@1)	42.4	48.2	45.2	43.6	40.6	28.9	40.5
DROP (3-shot F1)	83.7	88.3	83.9	71.2	85.5	77.0	78.7
MMLU-Pro (EM)	72.6	78.0	80.3	52.0	68.8	53.5	58.6
IF-Eval (Prompt Strict)	84.3	86.5	84.8	40.4	78.3	60.5	61.0
Mathematics							
MATH500 (Pass@1)	74.6	78.3	90.0	90.6	93.9	92.8	95.8
AIME 2024 (Pass@1)	9.3	16.0	63.6	50.0	69.7	55.5	68.2
AIME 2025 (Pass@1)	11.6	7.4	50.7	32.4	48.2	38.8	55.4
Code							
LiveCodeBench v5 (Pass@1)	32.9	38.9	53.8	41.9	53.1	37.6	57.8
LiveCodeBench v6 (Pass@1)	30.9	37.2	46.8	39.1	31.9	23.9	49.3

Table 4 Comparison between MiMo-7B-RL and other representative models.

efficient inference for MTP-equipped architectures.

Better Robustness In verl, vLLM is deployed using the *external launch* mode, which may show instability in some scenarios. We’ve enhanced engine robustness to address these issues. We clear computed blocks in *prefix caching* during pre-emption to maintain KVCache consistency. We disable asynchronous output processing when increasing the number of scheduler steps to ensure compatibility and optimize performance.

3.5 Post-Training Evaluation

3.5.1 Evaluation Setup

We comprehensively evaluate reasoning models across a diverse range of benchmarks:

Language understanding and reasoning: MMLU-Pro (Wang et al., 2024).

Scientific question answering: GPQA Diamond (Rein et al., 2024) with averaged score of 8 repetitions; SuperGPQA (Du et al., 2025).

Instruction following: IFEval (Zhou et al., 2023) with averaged score of 8 repetitions.

Reading comprehension: DROP (Dua et al., 2019).

Mathematics reasoning: MATH500 (Lightman et al., 2024); AIME 2024 (MAA, 2024) and AIME 2025 (MAA, 2025) with averaged score of 32 repetitions.

Coding: LiveCodeBench v5 (20240801-20250201) (Jain et al., 2024) and LiveCodeBench v6 (20250201-20250501) (Jain et al., 2024) with averaged score of 8 repetitions.

During evaluation, we set the sampling temperature to 0.6 and top-p to 0.95 for all benchmarks. We set the maximum generation length to 32,768 tokens for mathematics reasoning, coding, and scientific question answering benchmarks, and to 8,192 tokens for other benchmarks.

We compare MiMo-7B-RL against several strong baselines, including two non-reasoning models GPT-4o-0513, Claude-Sonnet-3.5-1022, and reasoning models OpenAI-o1-mini, QwQ-32B-Preview, DeepSeek-R1-Distill-Qwen-14B, and DeepSeek-R1-Distill-Qwen-7B.

基准	GPT-4o Claude-3.5- OpenAI QwQ-32B R1-蒸馏- R1-蒸馏- MiMo-0513 十四行诗-1022 o1-迷你预览 Qwen-14B Qwen-7B 7B-RL																			
常规																				
GPQA 钻石	49.9	65.0	60.0	54.5	59.1	49.1	54.4	超级 GPQA	42.4	48.2	45.2	43.6	40.6	28.9	40.5	下降	83.7	88.3	83.9	71.2
85.5	77.0	78.7	MMLU-Pro	72.6	78.0	80.3	52.0	68.8	53.5	58.6										
IF-Eval (Prompt Strict)				84.3	86.5				84.8	40.4				78.3	60.5	61.0				
数学																				
MATH500	74.6	78.3	90.0	90.6	93.9	92.8	95.8	AIME 2024	9.3	16.0	63.6	50.0	69.7	55.5	68.2	AIME 2025	11.6	7.4	50.7	
32.4	48.2	38.8	55.4																	
法典																				
LiveCodeBench v5	32.9	38.9	53.8	41.9	53.1	37.6	57.8	LiveCodeBench v6	30.9	37.2	46.8	39.1	31.9	23.9	49.3					

表 4 MiMo-7B-RL 与其他代表性模型的比较。

为配备 MTP 的架构实现高效推理。

更好的健壮性在 verl 中，vLLM 使用外部启动模式部署，在某些情况下可能会出现不稳定。我们增强了引擎的稳健性来解决这些问题。我们在抢占期间清除前缀缓存中的计算块，以保持 KVCache 的一致性。在增加计划程序步骤的数量时，我们会禁用异步输出处理，以确保兼容性并优化性能。

3.5 训练后评估

3.5.1 评估设置

我们根据各种基准全面评估推理模型：

语言理解和推理：MMLU-Pro (Wang et al., 2024) 。

科学问答：GPQA Diamond (Rein 等人, 2024 年) ，平均成绩为 8 次重复;SuperGPQA (Du et al., 2025) 。

说明如下：IFEval (周 et al., 2023) ，平均分数为 8 次重复。

阅读理解：DROP (Dua 等人, 2019 年) 。

数学推理：MATH500 (Lightman et al., 2024) ;AIME 2024 (MAA, 2024) 和 AIME 2025 (MAA, 2025) 的平均分数为 32 次重复。

编码：LiveCodeBench v5 (20240801-20250201) (Jain et al., 2024) 和 LiveCodeBench v6 (20250201-20250501) (Jain et al., 2024) ，平均分数为 8 次重复。在评估过程中，我们将所有基准测试的采样温度设置为 0.6，top-p 设置为 0.95。对于数学推理、编码和科学问答基准测试，我们将最大生成长度设置为 32768 个令牌，对于其他基准测试，我们将最大生成长度设置为 8192 个令牌。

我们将 MiMo-7B-RL 与几个强大的基线进行了比较，包括两个非推理模型 GPT-4o-0513、Claude-Sonnet-3.5-1022，以及推理模型 OpenAI-o1-mini、QwQ-32BPreview、DeepSeek-R1-Distill-Qwen-14B 和 DeepSeek-R1-Distill-Qwen-7B。

3.5.2 Evaluation Results

Table 4 shows the evaluation results. In mathematics reasoning, MiMo-7B-RL achieves top-tier performance among models of comparable parameter sizes, trailing only slightly behind DeepSeek-R1-Distill-Qwen-14B on AIME 2024. For algorithm code generation tasks, MiMo-7B-RL demonstrates extremely impressively results. On LiveCodeBench v5, it significantly outperforms OpenAI o1-mini, while on the latest LiveCodeBench v6, our model achieves a score of 49.3%, surpassing QwQ-32B-Preview by over 10 points, demonstrating its robust and stable capabilities. Notably, MiMo-7B-RL also maintains strong general performance, exceeding both QwQ-32B-Preview and DeepSeek-R1-Distill-Qwen-7B, though we only include mathematics and code problems for RL.

We also presents the evaluation results for different version of MiMo-7B in Table 5. MiMo-7B-RL-Zero is trained from MiMo-7B-Base, while MiMo-7B-RL is trained from MiMo-7B-SFT. As shown, RL from the base model exhibits a stronger growth trend, improving from 32.9% to on AIME 2024 for instance. Nonetheless, RL training from the SFT model achieves a higher performance ceiling, attaining the best results across all evaluated benchmarks.

Benchmark	MiMo-7B-Base	MiMo-7B-RL-Zero	MiMo-7B-SFT	MiMo-7B-RL
Mathematics				
MATH500	37.4	93.6	93.0	95.8
AIME 2024	32.9	56.4	58.7	68.2
AIME 2025	24.3	46.3	44.3	55.4
Code				
LiveCodeBench v5	32.9	49.1	52.3	57.8
LiveCodeBench v6	29.1	42.9	45.5	49.3

Table 5 Evaluation results of MiMo-Series models on mathematics and coding benchmarks

3.6 Discussion

In this section, we share insights and observations from our exploration of MiMo-7B’s post-training process, which we hope will benefit the research community.

SFT for Format Alignment In the initial RL training steps from MiMo-7B-Base, we observe that the model primarily learns to adapt the answer extraction function, e.g., “\boxed{ }” for mathematics problems. Therefore, we investigate a “light-weight” SFT to help the base model align with the expected answer format. However, as Figure 7 demonstrates, the resulting MiMo-7B-RL-LiteSFT model fails in both reasoning potential and final performance. While MiMo-7B-RL-LiteSFT begins with a higher performance than MiMo-7B-RL-Zero, it falls behind the base model’s trajectory after just 500 steps. Furthermore, when compared to MiMo-7B-RL, which undergoes “heavier” SFT, MiMo-7B-RL-LiteSFT exhibits a similar growth trend but significantly

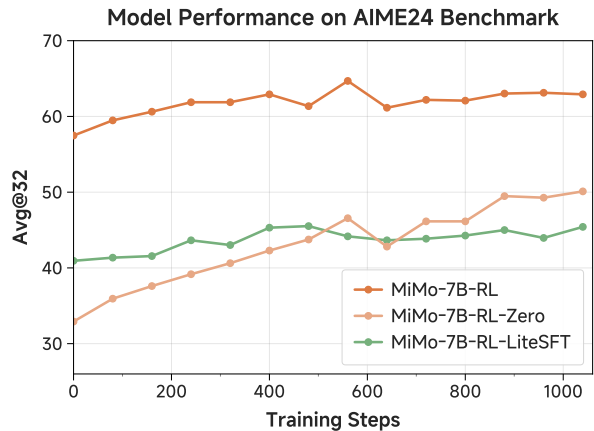


Figure 7 Performance comparison of three MiMo model variants during the RL process.

3.5.2 评估结果

评估结果如表 4 所示。在数学推理方面，MiMo-7B-RL 在同等参数大小的模型中取得了顶级性能，在 AIME 2024 上仅略微落后于 DeepSeekR1-Distill-Qwen-14B。对于算法代码生成任务，MiMo-7B-RL 展示了非常令人印象深刻的结果。在 LiveCodeBench v5 上，其性能明显优于 OpenAI o1-mini，而在最新的 LiveCodeBench v6 上，我们的模型获得了 49.3% 的分数，比 QwQ-32B-Preview 高出 10 分以上，展示了其稳健可靠的能力。值得注意的是，MiMo-7B-RL 还保持了强大的总体性能，超过了 QwQ-32B-Preview 和 DeepSeek-R1-Distill-Qwen-7B，尽管我们只包括 RL 的数学和代码问题。

我们还在表 5 中提供了不同版本的 MiMo-7B 的评估结果。MiMo-7B-RLZero 是从 MiMo-7B-Base 训练的，而 MiMo-7B-RL 是从 MiMo-7B-SFT 训练的。如图所示，基础模型的 RL 表现出更强劲的增长趋势，从 32.9% 提高到 AIME 2024。尽管如此，来自 SFT 模型的 RL 训练实现了更高的性能上限，在所有评估的基准中获得了最佳结果。

基准	MiMo-7B-Base	MiMo-7B-RL-Zero	MiMo-7B-SFT	MiMo-7B-RL
数学				
MATH500	37.4	93.6	93.0	95.8
爱 2024	32.9	56.4	58.7	68.2
爱 2025	24.3	46.3	44.3	55.4
法典				
LiveCodeBench 版本 5	32.9	49.1	52.3	57.8
LiveCodeBench 版本 v6	29.1	42.9	45.5	49.3

表 5 MiMo 系列模型在数学和编码基准上的评估结果

3.6 讨论

在本节中，我们分享了我们对于 MiMo-7B 训练后过程探索的见解和观察，我们希望这将使研究界受益。

用于格式对齐的 SFT 在 MiMo-7B-Base 的初始 RL 训练步骤中，我们观察到该模型主要学习调整答案提取函数，例如，“`\boxed{}`”用于数学问题。因此，我们研究了“轻量级”SFT，以帮助基本模型与预期的答案格式保持一致。然而，如图 7 所示，得到的 MiMo-7B-RL-LiteSFT 模型在推理电位和最终性能方面都失败了。虽然 MiMo-7B-RL-LiteSFT 的性能高于 MiMo7B-RL-Zero，但它落后于基本型号轨迹。此外，与经历“更重”SFT 的 MiMo-7B-RL 相比，MiMo-7B-RL-LiteSFT 表现出相似的增长趋势，但显着增长

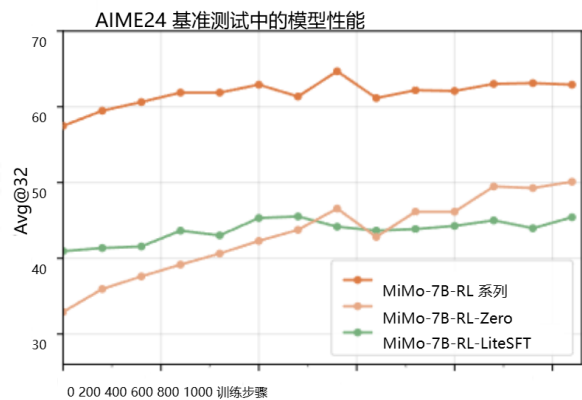


图 7 RL 过程中三种 MiMo 模型变体的性能比较。

underperforms due to its inferior starting point, ultimately leading to poorer final results.

Interference Between Different Domains During the later stages of RL training from MiMo-7B-Base, maintaining a performance balance between mathematics and coding tasks proves challenging. Between training steps 2000 and 2500, the model exhibits continuous improvement on code problems, while its performance on mathematical reasoning tasks fluctuates and declines. In contrast, RL training on the cold-started SFT model shows consistent improvements across both domains. Analysis of the model outputs reveals that the base model, with its strong exploration capabilities, tends to hack the reward for mathematics problems. For code problems, however, the test-case-based verifier makes reward exploitation significantly harder. This highlights the critical need for high-quality mathematical problem sets to ensure robust RL training.

Language Mixing Penalty Like DeepSeek-R1-Zero, we also observe language mixing issues during RL training on MiMo-7B-Base. To mitigate this problem, we introduce a language mixing penalty into the reward function. However, we find designing such a penalty function is challenging. While detecting Chinese characters in English responses is straightforward, the reverse is far more difficult, since mathematical equations and code inherently contain English words. As a result, the penalty not only fails to fully resolve language mixing but also introduces the risk of reward hacking, such as always generating English responses regardless of the question language.

4 Conclusion

This work introduces MiMo-7B, a series of LLMs which unlock advanced reasoning capabilities through optimized pre-training and post-training process. Exposed to diverse reasoning patterns during pre-training, MiMo-7B-Base possesses exceptional reasoning potential, outperforming models of significantly larger scale. For post-training, with our robust and efficient RL frameworks, we trained MiMo-7B-RL-Zero and MiMo-7B-RL which demonstrate superior reasoning capabilities across mathematics, code and general tasks. Notably, MiMo-7B-RL achieves 49.3% on LiveCodeBench v6 and 55.4% on AIME 2025, surpassing OpenAI’s o1-mini. We hope this work offers insights for developing more powerful reasoning models.

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MiMo-7B-RL-LiteSFT 表现出类似的增长趋势，但由于起点较差，表现明显不佳，最终导致最终结果较差。

不同域之间的干扰 在 MiMo7B-Base 的 RL 训练后期，在数学和编码任务之间保持性能平衡被证明是具有挑战性的。在训练步骤 2000 和 2500 之间，该模型在代码问题上表现出持续改进，而其在数学推理任务上的性能则波动和下降。相比之下，冷启动 SFT 模型的 RL 训练在两个领域显示出一致的改进。对模型输出的分析表明，基础模型具有强大的探索能力，往往会破解数学问题的奖励。但是，对于代码问题，基于测试用例的验证程序会使奖励利用变得更加困难。这凸显了对高质量数学问题集的迫切需求，以确保稳健的 RL 训练。

语言混合惩罚 与 DeepSeek-R1-Zero 一样，我们也在 MiMo-7B-Base 的 RL 训练期间观察到语言混合问题。为了缓解这个问题，我们在 reward 函数中引入了语言混合惩罚。但是，我们发现设计这样的惩罚函数具有挑战性。虽然在英文回答中检测中文字符很简单，但反之要困难得多，因为数学方程式和代码本身就包含英文单词。因此，这种处罚不仅无法完全解决语言混合问题，而且还引入了奖励黑客攻击的风险，例如无论问题语言如何，总是生成英语回答。

4 结论

这项工作介绍了 MiMo-7B，这是一系列 LLM，通过优化的训练前和训练后过程解锁高级推理能力。在预训练期间暴露于不同的推理模式，MiMo-7B-Base 具有非凡的推理潜力，性能优于明显更大规模的模型。在训练后，我们利用我们强大而高效的 RL 框架，训练了 MiMo-7B-RL-Zero 和 MiMo-7B-RL，它们在数学、代码和一般任务中表现出卓越的推理能力。值得注意的是，MiMo-7B-RL 在 LiveCodeBench v6 上实现了 49.3%，在 AIME 2025 上实现了 55.4%，超过了 OpenAI 的 o1-mini。我们希望这项工作为开发更强大的推理模型提供见解。

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